When Representation Change Misleads: Evaluating Encoder Adaptation in Time-Series Foundation Models

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Abstract

Fine-tuning a time-series foundation model changes its internal representations, and that change is often read as forgetting: drops in representation similarity are used to locate and predict it, and freezing, weight anchoring and low-rank adapters are motivated partly by preserving the pre-trained representation. We test that reading against the intervention it stands in for, fine-tuning with and without the encoder frozen on the same data and scored on held-out windows. Across **32 screened cells** (Moirai-S/B/L, Chronos-T5, TimesFM-2.5; six datasets; two horizons), centered kernel alignment (CKA) does not track the intervention. Within one backbone the least drifted cells are the only ones that degrade; a pre-registered positive control drives CKA to 0.063 while the targeted capability improves by 23.3%; PENDING: out-of-domain retention sentence; and out of sample, adding CKA to backbone and protocol lowers predictive R^2 from $+0.201$ to $+0.176$. Two measurement choices decide such comparisons: scoring on the windows used for selection reverses the intervention's sign in 4 of 31 cells, and an unfitted linear baseline puts 16 of 21 Moirai cells on the wrong side of a capability screen that only 7 of 31 pass against a fitted one. No cell meets a prespecified definition of harmful encoder adaptation. The intervention costs one extra training run, and we recommend running it rather than reading drift.

1 Introduction

Fine-tuning a time-series foundation model reshapes its internal representations, and that reshaping is often read as forgetting: layer-wise CKA drops are used to locate where forgetting happens (Ramasesh et al., 2021), representation similarity to predict how much a foundation model will forget (Mian et al., 2026), and feature distortion to explain why fine-tuning underperforms out of distribution (Kumar et al., 2022). The same reading motivates freezing the encoder, regularising toward the pre-trained weights (Li et al., 2018; Kirkpatrick et al., 2017), and low-rank adapters, which time-series work adopts partly to retain the pre-trained representation (Gupta et al., 2024; Laglil et al., 2026) (and partly for efficiency or against overfitting). Insofar as they protect the representation, those prescriptions presuppose that it was worth preserving *on the target task*.

Whether a change was harmful is an interventional question: fine-tune the whole model, fine-tune it again with the encoder frozen on the same data and budget, and compare the two on windows neither run selected on. That costs one extra training run, so we do not offer CKA as a cheaper stand-in for it. We ask the converse: when an analysis reads a CKA drop as evidence of forgetting, is that reading warranted for time-series foundation models? Tested against the intervention, it is not.

One checkpoint, two datasets, opposite lessons. Fine-tune Moirai-Base on ETTh1 and its encoder barely moves (CKA 0.755 against the pre-trained encoder), and the model ends 22.8% *worse* than not fine-tuning at all. Fine-tune the same checkpoint on ETTh2 and the encoder restructures far more, to CKA 0.460, and the model ends 31.4% *better*. Read drift as damage and you would have protected the run that needed no protection and left alone the one that did.

What we test, and on what. We screen **32 cells**: three backbones (Moirai Small 13.8M, Base 91M, Large 311M; Chronos-T5 (Ansari et al., 2024); TimesFM-2.5 (Das et al., 2024), with MOMENT (Goswami et al., 2024) screened and excluded), six datasets and two forecast horizons. Each

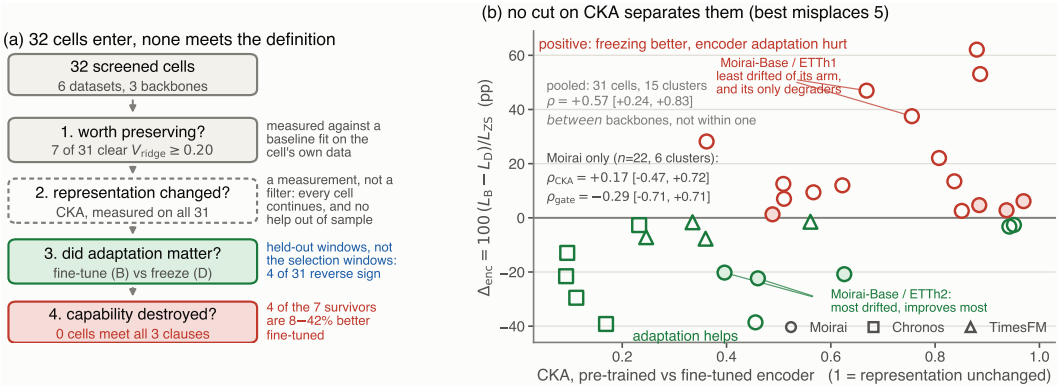


Figure 1: **(a)** The screen of §3 over the 31 cells that carry a selection-split gate: CKA filters none of them, and no cell meets the degradation definition. **(b)** The same cells as CKA against held-out Δ_{enc} (positive = freezing better): no cut on CKA separates the cells adaptation helped from the cells it hurt. The labelled pairs are the running example: Moirai-Base's two *least* drifted cells are its only two that degrade, and its two most drifted improve the most (§6). Filled markers clear the gate; both ρ are clustered by backbone×dataset, and only Moirai has enough cells to estimate one within a backbone (Appendix A16).

cell pairs a full fine-tune with a frozen-encoder run, and on the Moirai cells with a strict freeze under which nothing upstream of the forecasting head can move, at 3–10 seeds. We then ask four questions in order (Figure 1a). Is there a pre-trained capability worth preserving (§4)? Does letting the encoder adapt help or hurt (§5)? Does CKA order the answer (§6)? And where damage is forced, or looked for on other datasets, does CKA see it (§7)? Every claim is about the checkpoints, datasets and protocols we evaluated; the readings we decline are listed under *Scope of the claims* in the Ethics Statement.

Contributions.

1. **Drift does not track the intervention.** Within Moirai-Base the least drifted cells are the only ones that degrade; a pre-registered positive control drives CKA on the original task's inputs to 0.063 while that task *improves* by 23.3%; and PENDING: out-of-domain retention result. Across cells, within Moirai, the only backbone with enough cells to ask, ρ=+0.168 (clustered CI $[-0.469, +0.717]$): an interval that rules out a strong positive association but not a moderate one, so we claim no ordering rather than the absence of one. Out of sample, adding CKA to backbone and protocol lowers leave-one-cluster-out R^2 from +0.201 to +0.176 (§6).
2. **Two measurement prerequisites for any such comparison.** Scoring the frozen-encoder comparison on the windows it was selected on reverses its sign on 4 of 31 cells, all toward flattering the frozen encoder; and an unfitted linear baseline would move 17 of Moirai's 21 cells across the capability screen. With both in place, **only 7 of the 31 scored cells** show a pre-trained advantage over a linear regression fit on the target data, all of them Moirai on ETTh2 and ETTm2 (§§3, 4).
3. **No harmful encoder adaptation under a prespecified definition.** On multiplicity-adjusted paired intervals the 31 cells split 2 freezing-better, 4 adaptation-better and 25 inconclusive, and both freezing-better cells are ones the screen rejects; on five of the seven that pass it, full fine-tuning improves on zero-shot (§5).

2 Background and Related Work

Related work: three lines, and the gap between them. *(i) Forgetting* is studied in continual learning, NLP and fine-tuning dynamics (Kirkpatrick et al., 2017; Li & Hoiem, 2018; Luo et al., 2023; Kumar et al., 2022; Andreassen et al., 2022), including sequential fine-tuning of TSFMs (Karaouli et al., 2025), and supplies the prescriptions we test. **This is not a continual-learning study**: no task sequence, no loss on a prior task, one adaptation step, so "forgetting" here is degradation relative to the same checkpoint's zero-shot on the same data, and a catastrophic-forgetting result and ours can

hold at once. *(ii) TSFM adaptation* recipes span probabilistic fine-tuning, generative pre-training, LLM reprogramming and cross-task transfer (Rasul et al., 2024; Liu et al., 2024; Jin et al., 2024; Zhou et al., 2023), and are evaluated on target-task accuracy. *(iii) Representation analysis* supplies the measurement (representation similarity and probing (Kornblith et al., 2019; Hewitt & Manning, 2019; Pimentel et al., 2020)), including work probing frozen TSFM internals directly (Wiliński et al., 2025), where we instead track how representations *change under fine-tuning*. CKA has been used to localise forgetting (Ramasesh et al., 2021), and in vision classification a linear probe shows that a large CKA drop can coexist with little loss of knowledge (Davari et al., 2022); we ask the same question of TSFMs, against an intervention rather than a probe. **None of the three establishes that the pre-trained representation was worth preserving on the target task**, which is the premise (i)'s prescriptions rest on and what this paper measures.

Pre-training value gate. The gate is deliberately cheap and *downstream-only*: the practitioner runs the frozen model zero-shot on the target set and compares against a lookback-96 linear regression *fit on that same dataset* (§3), so neither pre-training data nor pre-training compute is required. We report V_b as a continuous quantity for every cell (Table 5) rather than only its verdict, and the 0.20 operating point is calibrated to ETT short-horizon benchmarks, where linear baselines are famously competitive with deep forecasters (Zeng et al., 2023). That competitiveness is the whole reason the gate has teeth: **7 of the 31 scored cells clear it**, all on Moirai and confined to two series, and in the other 24 the pre-trained model has no measured advantage for fine-tuning to destroy.

Task-level consequence: forgetting. We define forgetting as signed relative change in zero-shot-canonical task metric on held-out test: forg.$\% = 100 \cdot (M_{\text{FT}} - M_{\text{ZS}})/|M_{\text{ZS}}|$ (MSE for forecasting, so forg.>0 means the fine-tuned model is worse than the zero-shot one: *task-level forgetting*).

Decoupling harmful forgetting from benign specialisation. Representational change that redirects features toward the target (*benign specialisation*) must be separated from change that discards generally useful structure (*harmful forgetting*). We separate them by *value-gate conditioning*: change can be harmful only where the structure it discards was valuable, i.e. in a gate-pass cell; in a gate-fail cell the same change is benign by construction, because no demonstrated advantage existed to lose. Gate-passing plus forgetting>0 is therefore necessary for harm but not sufficient, and whether the encoder's *change* carried the loss is settled by intervention (full fine-tuning against a frozen encoder on the same data for the same epochs), and not by observing the representation.

3 Protocol and Definitions

Models and training. Moirai (Woo et al., 2024) predicts a probabilistic distribution over the horizon; we report MSE on the median of 20 forecast samples at h=96 and h=192 (Appendix S2). Moirai is trained with AdamW (η=10^{-4}, weight decay 10^{-4}), batch size 16 and gradient clipping at 1.0, for 20 epochs with no early stopping: the 21 Moirai cells on ETT, Weather and Electricity are read at their final epoch. The other trainers select an epoch on the selection windows (never the held-out ones): Moirai/ILI and TimesFM restore their best epoch under patience 3 and 7, and Chronos keeps its best of 30 epochs under full fine-tuning and trains its frozen-feature head to convergence (Table 4). **22 of the 31** cells train on 1k windows, and every table prints each cell's own n. Reproducing a Moirai fine-tune requires the `uni2ts` scaler patch of Appendix S1.

Three splits. The *training* split fits the fine-tuning updates and every baseline; the *selection* split is where any checkpoint is chosen and where the capability screen is scored; the *held-out* split is disjoint from both and is where every outcome is scored. A cell's admission therefore never uses the windows that later judge it.

Capability screen. A checkpoint can only have a capability worth preserving relative to something a practitioner could deploy instead, so the baseline is part of the quantity: for a baseline b, $V_b = 1 - \text{MSE}_{\text{ZS}}/\text{MSE}_b$, positive when the zero-shot (ZS) model beats b. We threshold it at 0.20, a prespecified operating point and not an established boundary, and call the result the *gate*. Our primary b is one ridge map $\mathbb{R}^{96\times F} \to \mathbb{R}^{h\times F}$ (λ=10^{-4}) fit by least squares on the training windows and applied unchanged to the selection windows. It sees 96 steps of context against the pre-trained model's $96{+}h$, which if anything favours the pre-trained model; §4 also scores a ladder of eight baselines.

Conditions. Full fine-tuning (B) updates every parameter. The frozen encoder (D) freezes the encoder stack but leaves the input projection and mask encoding trainable, so its CKA is near but not exactly 1 on Moirai (0.889–0.9999). The strict freeze (H) also freezes those two modules, so only the output head trains and the encoder's output is fixed by construction (CKA exactly 1.000). LoRA (r=8), L2-SP and EWC are reported as a mitigation spectrum in Appendix A6. Figure 4 shows which modules move under each.

The estimand. For a cell with S paired seeds and a frozen arm F,

$$\Delta_{\text{enc}} = \frac{1}{S}\sum_{s=1}^{S} 100 \cdot \frac{\mathcal{L}_{\text{test}}(\text{B}_s) - \mathcal{L}_{\text{test}}(\text{F}_s)}{\mathcal{L}_{\text{test}}(\text{ZS})}, \tag{1}$$

positive when freezing does better. The shared zero-shot denominator only rescales each cell so that different loss scales are comparable. On Moirai, where every cell has both controls, we lead with F=H, the cleaner contrast, and report F=D beside it; across backbones we use F=D, the one control all three share. The two disagree in sign on PENDING: N of 22 Moirai cells (§5). Δ_{enc} is the contrast the design *manipulates*. CKA is a *post-treatment* measurement, taken on the object the treatment changed, so it can only be a candidate predictor of Δ_{enc}, and that is the role we test.

Representation diagnostics. Linear CKA (Kornblith et al., 2019) between pre-trained and fine-tuned representations, and weight drift (ℓ_2 distance from the pre-trained parameters). Every CKA in the body is of the final encoder output, mean-pooled over tokens, on a fixed prefix of the selection split: the comparison a practitioner holding two checkpoints can make without choosing a layer or token position. Token-flattened and per-layer variants and a random-initialisation floor are in Appendices A10 and A9.

Definition (degradation of a demonstrated capability). A cell is a degradation when (i) $V_b \geq 0.20$ on selection windows, (ii) full fine-tuning ends worse than zero-shot, and (iii) the frozen arm ends better than zero-shot, which localises the loss to encoder updates rather than to the data or the epoch budget. Clauses (ii) and (iii) are tested as paired 95% intervals on B−ZS and F−ZS, propagating the zero-shot reference's error and adjusted across cells by Benjamini–Hochberg (Appendix A14), and also in the pre-registered form in which every seed must agree. **Both return the same count**. Every ± in the body is a standard error over seeds.

Two measurement choices that decide this result. *(1) Score the frozen-encoder comparison on windows you did not select on.* Scoring Δ_{enc} on the selection windows instead reverses its sign on 4 of our 31 cells, and all four reverse toward flattering the frozen encoder (§5.3). *(2) Fit the linear baseline.* An unfitted per-window trend extrapolation, a denominator a constant predictor beats, would move 17 of Moirai's 21 cells across the threshold, 16 of them to gate-pass, and so reverse the conclusion (Appendix A12).

4 Experiment 1: Is There a Capability Worth Preserving?

A claim that fine-tuning damages valuable pre-trained features needs a cell where such features are demonstrably present. Across **32 screened cells** (21 Moirai: three capacities, five datasets, two horizons; 5 Chronos-T5, 5 TimesFM-2.5 and 1 Moirai/ILI), seven clear the gate, and all seven are Moirai: five on ETTh2 and two on ETTm2. The gate is scored on selection windows, which exist for 31 of the 32 (TimesFM/Electricity has no paired frozen-encoder run, so it is screened but carries neither a gate nor an intervention row), and every denominator below is 31.

Moirai: 7 of 21 (Table 5). ETTh2 clears at every capacity: Small h=96 (+0.874) and h=192 (+0.932), Base h=96 (+0.777) and h=192 (+0.925), Large h=96 (+0.875). ETTm2 clears at two of four, Small h=192 (+0.352) and Base h=192 (+0.211). Every other dataset fails at every capacity, and every failing cell is below the baseline rather than short of the threshold. Capacity does not help: Moirai-Base is 6.6× larger than Moirai-Small and further behind the baseline on both ETTh1 and Weather.

Chronos-T5-Small: 0 of 5; TimesFM-2.5: 0 of 4 (Table 9). All nine fall in a narrow band about zero, −0.275 to +0.082. On ETTh2, where Moirai clears at every capacity, Chronos is below the fitted baseline (−0.184 on selection, −0.325 on held-out), so the advantage is not a property of the dataset alone.

One rung of eight. Scored against a ladder of eight baselines on the same windows, with the training mean as an admissibility floor, **no cell of the 31 clears** 0.20 against the strongest admissible rung it faces, and none does on two further window sets. **20 of the 31 clear it against at least one admissible rung (14 Moirai, 3 Chronos, 3 TimesFM)**, and that union is the scope of every drift analysis below, so those analyses span all three backbones. "Seven cells clear the gate" always means seven clear the fitted ridge rung; per-rung counts, a matched-lookback baseline and the four cells a near-constant rung would admit are in Appendix A11.

Prospectively, the gate is a conditioning device, not a detector. Predictions for eight unrun cells were committed to git before any of them was trained. None of the eight degraded and the gate flagged all eight (**TP 0, FP 8, FN 0**), a test at zero prevalence that shows the gate's positive predictions were wrong here and says nothing about its sensitivity. A second registered batch, on 9 of its 10 cells, gives specificity 0.78 (Appendices S3, S4). What the gate does is decide where the harm question can be asked at all: a cell that fails it has no demonstrated advantage to lose, and 24 of our 31 fail it.

5 Experiment 2: Does Encoder Adaptation Help or Harm?

5.1 What Fine-Tuning Does to the Seven Survivors

The seven cells Experiment 1 admits are the whole evidence base for a claim about damage to valuable features, and they do not support one (Table 17).

On four of the seven, full fine-tuning improves the checkpoint in every seed. On Moirai-Base at h=96 and h=192 and Moirai-Large at h=96, all on ETTh2, forgetting is -31.4%, -36.3% and -42.4% against zero-shot, negative in 3/3 seeds each, at CKA 0.460, 0.396 and 0.626, the largest representational change among the seven. Freezing the encoder captures only part of that gain (-9.1%, -16.1%, -21.6%), so most of the improvement comes from encoder adaptation. The fourth, Moirai-Base/ETTm2 at h=192, improves by 8.4%, and the frozen encoder does slightly better there (-9.7%).

On the remaining three, the mean harm does not survive inspection. Moirai-Small/ETTh2 at h=192 reads $\text{forg.}_{\text{B}}=+1.99\pm3.32$ with 5 of 10 seeds positive, the closest the matrix comes to degradation. At h=96 the positive mean ($+9.71\pm10.58$, 4 of 10 seeds) comes from one seed that diverges under the frozen encoder too, an optimisation failure rather than an encoder effect; we keep it in every mean. Moirai-Small/ETTm2 at h=192 improves on average (1.10%) with 3 of 5 seeds harmful.

5.2 Where the Freezing Boundary Is Drawn

The ordinary frozen encoder (D) leaves the input projection and mask encoding trainable, so a "freezing wins" reading could come from those modules adapting rather than from the encoder staying fixed. We therefore ran the strict freeze (H) on every Moirai cell (PENDING: 22 cells, 3–10 seeds; Table 11). B−H and B−D agree in sign on PENDING: N cells. They disagree on PENDING: the four Moirai/ETTm2 cells and any of the six new ones, and on every disagreeing cell the strict control moves toward adaptation: on ETTm2, Moirai-Small h=96 goes from B−D $+7.3\pm2.9$ to B−H -6.8 ± 2.0 and Moirai-Base h=96 from $+7.0\pm2.4$ to -9.2 ± 3.1, so the apparent freezing advantage on that dataset was carried by the input projection. The strict freeze therefore recovers no degradation case; it removes the ones D appeared to suggest. All Moirai counts below are reported under both controls.

5.3 Held-Out Scoring Is a Precondition

Scoring Δ_{enc} on the selection windows instead of the held-out ones **reverses its sign in 4 of the 31 intervention cells**, all in the same direction: on the selection windows the frozen encoder wins, on held-out windows full fine-tuning does (Table 8). Two of the four are among the seven gate-passing cells (Moirai-Base and Moirai-Large on ETTh2, which move from $+3.8$ and $+7.2$ to -22.3 and -20.8). A frozen-versus-full comparison scored on the windows a checkpoint was selected on is therefore not a weaker version of the held-out one; it can give the opposite answer.

5.4 No Degradation Instance Survives a Prespecified Definition

The three-clause definition of §3 returns **zero cells** of the 31, under both the interval and the registered form and, on Moirai, under both controls. Clause (i) removes 24 outright: they have no measured pre-trained advantage, so a worse score after fine-tuning replaces nothing demonstrated. Of the seven that remain, four fail clause (ii) with every seed agreeing, and the other three fail it cell by cell (§5.1). Relaxing the definition does not manufacture a case: across 8 rungs, at either gate threshold, **the largest count anywhere is 2**, reached only when the means clause is dropped altogether (Appendix S8).

Per-cell inference. Paired 95% intervals on Δ_{enc} (F=D), Benjamini–Hochberg adjusted across the 31 cells, call **2 cells freezing-better, 4 adaptation-better and 25 inconclusive**; under F=H on Moirai, PENDING: counts. The two freezing-better cells are Moirai-Small/ETTh1 at both horizons, which the screen rejects, while two of the four adaptation-better cells pass it, at $+0.925$ and $+0.875$. A pre-registered seed top-up adds a third freezing-better cell, which the screen admits (Moirai-Small/ETTm2 h=192, gate $+0.352$), and the definition still returns zero (Appendix A14). Most calls are inconclusive because the design is underpowered for small effects, and we say so rather than read them as null: the survivors' dispersion gives minimum detectable effects of 4.7–14.7 pp, and two survivors would need 182 and 212 paired seeds to resolve their observed effect. Over the 20 value-cells, Δ_{enc}=$+0.7$ with clustered CI $[-11.1, +14.1]$. Changing the gate's estimator does admit cells (persistence admits three and GBM one), but all four rest on a denominator within 5% of the training-mean floor, so a weaker baseline is what manufactures them (Appendix A11).

We do not conclude that fine-tuning never damages a pre-trained capability, only that on these benchmarks we could not find one instance that meets its own definition: no cell's advantage survives every admissible rung of the ladder, and where the fitted-ridge advantage exists, fine-tuning on 500–1,000 samples usually improves on it.

6 Experiment 3: Does Drift Order the Outcome?

CKA is computed after fine-tuning, on the model fine-tuning changed, so it cannot replace the intervention as a decision rule. It is worth testing anyway because that is how it is used: analyses of fine-tuned models report a CKA drop as evidence of forgetting (§1). The question is whether such a drop carries information about the outcome the intervention measures. We test it in three ways, from the most direct to the most aggregate.

Counterexamples within one backbone. Within Moirai-Base the two *least* drifted cells are the only two that degrade (ETTh1 at CKA 0.755 and 0.669, +22.8% and +39.5%), while its two most drifted improve the most (ETTh2 at CKA 0.460 and 0.396, −31.4% and −36.3%; Table 22). On Moirai-Small/ETTh2, one of the seven survivors, sweeping $n \in \{500, \ldots, 10\text{k}\}$ drives CKA down monotonically from 0.949 to 0.518 while held-out forgetting *falls*, from +5.2% at 500 to $-18.5 \pm 2.8\%$ at 10k, with every seed improving on zero-shot from n=1k on: more drift, better outcome (Table 6, Figure 3). Inside one encoder, unfreezing all 6 layers rather than the top 3 lowers CKA from 0.716 to 0.465 and improves held-out forgetting from −9.5% to −22.9% (10 seeds each; Appendix A10). Across mitigation strategies on that cell, the three weight-anchoring arms span CKA 0.939–0.972 and land within 0.003 MSE of one another (Appendix A6). Each of these contradicts "lower CKA, more damage" as a rule; none of them establishes the reverse.

Other backbones. None of the nine Chronos and TimesFM cells passes the gate and their training protocols differ (Table 4), so we read them beside Moirai rather than pooling them. Four of the five Chronos-T5-Small cells restructure to CKA 0.092–0.169, at or below the 0.215 ± 0.054 of an *untrained* encoder of this architecture (Appendix A9), yet their outcomes span −41.4% on ETTm2 to +20.6% on ETTh1, and ETTh1, the least drifted, is the only one that degrades. The four TimesFM-2.5 cells sit at milder drift and all improve (Table 9).

Association across cells. Pooled over all 31 cells, CKA correlates with Δ_{enc} (ρ=$+0.567$, clustered CI $[+0.239, +0.828]$), but that is mostly the gap between backbones, which a practitioner knows before computing anything (Figure 1b). Within Moirai, the one backbone with enough cells, ρ=$+0.168$ (22 cells, 6 clusters, CI $[-0.469, +0.717]$; under F=H, PENDING: ρ and CI). The interval excludes a strong positive association but not a moderate one. This design would detect a true ρ of 0.50

with 80% power (Appendix A16), so an association of 0.5 or more would probably have shown; a weaker one may exist, and we claim only that CKA does not *establish* an ordering. Acting on CKA as the prescriptions do, freezing where drift is largest, is worth -6.97 pp $[-10, +14]$ against always adapting (Appendix S13), and ℓ_2 weight drift and five further geometric measures do no better (Appendices A15, S11).

Out-of-sample prediction. With the predictor ladder and the deciding statistic fixed in advance, leave-one-cluster-out R^2 over the 15 clusters rises from $+0.000$ to $+0.137$ on backbone identity and to $+0.201$ with horizon and $\log n$, then *falls* to $+0.176$ once CKA is added ($\Delta R^2 = -0.025$; no improvement in 69% of cluster-bootstrap draws). What is predictable about Δ_{enc} here is predictable from the protocol; CKA adds nothing measurable to it (Appendix A17).

7 Experiment 4: Does CKA See Damage Where It Can Occur?

Every cell above is fine-tuned and scored on the same series, so Experiments 2 and 3 measure what adaptation does to the target task. Forgetting proper is a loss of what the model could do elsewhere, and a diagnostic cannot be shown to miss damage that never occurred. This section looks for damage in the two places it can occur: on a task constructed to conflict with the pre-trained prior, and on datasets the model was not fine-tuned on. Both arms were pre-registered, with thresholds and outcomes committed before the first run.

7.1 A Positive Control: Near-Orthogonal Drift, and the Capability Survives

Task A is the strongest capability the screen admits (Moirai-Small/ETTh2, $h=192$, $V_{\text{ridge}}=+0.932$). Task B is a synthetic series, fixed by hash before any run, whose correct forecast is the *negation* of the seasonal-naive prior, task A's own strongest admissible baseline. We fine-tuned on B under lr $\in \{10^{-4}, 10^{-3}, 10^{-2}\}$ with both encoder conditions at 3 seeds and measured A (Appendix A18). Task B was learned in **21 of 21** runs (minimum $+56.6\%$ against a registered 20%). At the top rung CKA on task A's inputs falls to 0.063 and ℓ_2 drift reaches 2795, against 0.999 and 7.2 at the gentlest, and task A gets *better*, by $23.3 \pm 2.0\%$ in every seed; across the 21 runs CKA and retention correlate at $\rho = +0.466$, the wrong sign for a damage detector. The registered destruction threshold (retention loss $\geq 50\%$ in a majority of seeds) was never reached: the worst run lost 26.7%. A representation can be driven most of the way to orthogonality by training in conflict with a capability, and the capability survives.

7.2 Out-of-Domain Retention

We fine-tuned each of the seven gate-passing Moirai cells under full fine-tuning, the frozen encoder and the strict freeze (3 seeds each) and scored every final checkpoint zero-shot on the held-out windows of the four datasets it was not trained on, measuring CKA on those datasets' own inputs (Appendix A4). The registered threshold for harmful forgetting is a loss at least 10 pp larger under full fine-tuning than under the strict freeze, in every seed.

PENDING: Result: how many (cell, dataset) pairs meet it; retention range for B, D, H; Spearman between 1−CKA and retention with its cell-clustered interval; which registered o

8 Limitations

(1) *One backbone carries the within-backbone evidence.* The seven gate-passing cells and the within-backbone association are Moirai on two series; Chronos and TimesFM gate-fail everywhere and use different trainers, so they show the dissociation but do not replicate the analysis (§6). PENDING: second-family runs on gate-passing datasets, if completed. (2) *A null is not a proof of absence.* What would settle the harm question is a cell with a large pre-trained margin over a strong baseline, which these benchmarks do not supply (§4). (3) *Seed counts are small*, 3 to 10 per cell, so most per-cell calls are inconclusive by construction (§5.4). (4) *Horizon differs by backbone*: $h=24$ for Chronos and TimesFM, $h \in \{96, 192\}$ for Moirai, so the two are never pooled. (5) *Out-of-domain retention is measured on the paper's own benchmarks*, not on Moirai's pre-training corpus (§7.2). (6) *We measure whether restructuring was necessary, not what it changed*: linear probes on these encoders do not generalise under a temporal split (supplementary `ERRATA.md`). Further detail is in Appendix A2.

9 CONCLUSION

Representation drift is widely read as forgetting. Against the intervention it stands in for, on three time-series foundation models and six benchmarks, that reading does not hold: the least drifted cells can be the ones that degrade, a representation driven nearly orthogonal can retain its capability, and CKA adds nothing to what backbone and protocol already predict. We could not find one cell where fine-tuning demonstrably damaged a pre-trained capability, and most of our cells had no measurable capability to damage. None of this makes CKA a poor measure of similarity; it makes similarity a poor measure of harm.

The practical advice follows from the two measurement choices that decided our results. First check that the pre-trained model beats a baseline fitted on your own data, because a forgetting result on a cell with no pre-trained advantage is a statement about the benchmark. Then compare full fine-tuning against a strictly frozen encoder on windows neither run selected on. That comparison costs one extra training run, and it answers the question drift only appears to.

ETHICS STATEMENT

Data. Every experiment uses public benchmark data and involves no human subjects and no personally identifying information. The forecasting datasets (ETTh1, ETTh2, ETTm2, Weather and Electricity) are standard public releases of instrument and utility measurements. The ILI cell uses public aggregate influenza-surveillance counts, not individual health records. The IoT negative control uses N-BaIoT, network traffic captured from a laboratory testbed of consumer devices rather than from a deployed network. We introduce no new data collection.

The main risk in this paper is that its negative result is read too broadly. We report that on these benchmarks we could not construct a cell in which fine-tuning demonstrably damages a pre-trained capability, and that a diagnostic we proposed for predicting such damage has no positive instance to predict. That is a statement about six standard forecasting benchmarks and three backbones, and §8 states what it does not license. It is *not* evidence that fine-tuning is safe in general, and a practitioner who reads it as permission to skip the frozen-encoder control would be drawing the opposite of our recommendation: our prescription is to *run* that control, because we found the observational shortcuts unreliable. Domains where a pre-trained checkpoint holds a large, well-established advantage (which these benchmarks do not, on 24 of the 31 cells we screened) are exactly where the question remains open.

Scope of the claims: one adaptation protocol, not fine-tuning in general. Three facts bound every number in this paper, and we state them together here rather than leaving them to be assembled from three appendices. *(i) The training objective differs by backbone, and one of the three is not native.* Moirai is fine-tuned through its own probabilistic NLL, and TimesFM through its own point head's loss (MSE on the mean channel plus pinball on the nine decile channels), with no head attached to either. The five Chronos-T5-Small cells in the main matrix are instead fine-tuned through an *added* MSE regression head on mean-pooled encoder features, which bypasses the tokenised cross-entropy Chronos was pre-trained with. That mismatch is why Chronos's per-condition forgetting is flagged confounded in the code and excluded wherever a forgetting comparison is made (Appendix S12), and why the M4-Monthly arm, which *is* fine-tuned through the native cross-entropy, is reported separately (Appendix S6) instead of pooled with those five. *(ii) The outcome is one metric on one channel.* Every forecasting result is MSE on a single target channel at a fixed horizon; we test no multivariate-target, multi-task or classification adaptation beyond the IoT negative control. *(iii) One adaptation step, not a task sequence* (§2). "Encoder adaptation" in this paper therefore means what this protocol does, and our results say nothing about a backbone fine-tuned through its own decoder on a task we did not run.

What we do not claim. Five readings of this paper's negative results are ones we explicitly decline. *(i)* That CKA fails to measure representation change: it does measure it, and we use it as a measurement throughout — what we test is whether that measurement substitutes for an intervention. *(ii)* That CKA can never predict downstream utility. We report that it does not reliably order the intervention outcome across the cells we ran, with the one backbone family large enough to condition on carrying the cleanest evidence; a larger matrix, another family or another protocol could show otherwise. *(iii)* That fine-tuning never causes catastrophic forgetting, or that fine-tuning is safe. We

could not construct a cell on these benchmarks in which it demonstrably destroys a demonstrated capability, which is a failure to demonstrate and not a demonstration of absence. *(iv)* That freezing, L2-SP, EWC or low-rank adaptation are useless. Our prescription is to *run* the frozen-encoder control rather than to skip it, because it is the observational shortcuts we found unreliable, not the mitigations. *(v)* That foundation models are useless for forecasting, or that these results establish the absence of encoder-adaptation effects in time-series foundation models in general: seven of 31 cells cleared a deliberately weak screen, and the domains where a pre-trained advantage is large and well established are exactly where the question stays open. What we do claim is narrower than any of those: **within the intervention cells we evaluated, and with the cleanest evidence inside the Moirai family, CKA is not an actionable substitute for the intervention, and no cell meets our prespecified criterion for harmful encoder adaptation.** Appendix A1 tabulates every claim we do make with its evidence and its strength.

The value gate is a screen, not a verdict on a checkpoint. A cell that fails the gate is one where the pre-trained model does not beat a linear regression fit on that cell's own target data; it is *not* a cell we claim holds no pre-trained knowledge, and V_b is a property of the (checkpoint, dataset, baseline) triple rather than of the checkpoint (§3). The screen's role is to decide where the question "was something valuable destroyed?" is even askable. Two things follow that are easy to misread. Failing cells here fail *widely*: every failing Moirai cell is below its baseline rather than short of the 0.20 threshold, so the emptiness of the screen is not an artefact of where the bar sits. And gate-failing cells are still measured and still reported — CKA, Δ_{enc} and the paired intervals are computed on all 31 — because the dissociation between drift and outcome is a claim about the measurement, not about the survivors.

Broader impact. This is a diagnostic study on public forecasting benchmarks, and the effect we intend is to raise the bar for claims about forgetting in time-series foundation models: a paper reporting that fine-tuning damaged pre-trained features should say what baseline established that the features were valuable, and how that baseline was fit. The controlled comparison remains the way to answer the question for any particular cell, and §5.4 claims only that the harm cannot be demonstrated on these benchmarks.

Corrections to an earlier version. An earlier public version of this work used an unfitted linear baseline for the capability screen and reported a linear-probe diagnostic that does not survive a temporal split. Both are corrected here, and the supplementary file `ERRATA.md` lists every changed claim.

Compute. The experiments are small by contemporary standards: no run exceeds 10,000 training windows, and the sweep reported here runs on a single GPU or on Apple silicon. We report the protocol deviations forced by the hardware available (Appendix A10) rather than presenting the runs as environment-independent.

Reproducibility Statement

The paper is organised so that every quantity can be traced to a stored run record, and we have tried to make the failure modes we hit ourselves cheap for others to avoid.

Where the numbers come from. Every table the source `\inputs` is written by a named emitter that reads the per-run JSON records (none is assembled by hand), and `scripts/rederive_all.sh` runs all of them in one command and then reports which generated files changed. A clean run means the PDF agrees with the run records; any diff is a number that was stale. The per-run records carry the fields the tables use (`final_cka`, `forgetting_pct`, `final_val_mse`, `final_weight_drift` and the early-stopping block), together with the seed, epoch budget and sample count, so the dispersion conventions can be recomputed rather than taken on trust.

What "reproducible" means here, exactly, and what it does not. Auditability is not independent reproduction, and we distinguish three tiers rather than letting the stronger word cover the weaker work. *(i)* Every table and figure is re-derivable from a clean clone with no data and no checkpoints, because the run records are in the release; that is what `rederive_all.sh` checks. *(ii)* Re-running the value gate or any fine-tuning cell additionally needs the benchmark CSVs, which are public but not ours to redistribute. The obstacle there is not the download but knowing you fetched the same file: ETT and Weather circulate in variants that differ in column order and date parsing, and a variant with the same name changes every window and so every MSE.

`results/data_manifest.json` therefore records the SHA-256, shape and date range of each CSV we used, and `scripts/data_manifest.py -check` verifies a local copy against it. *(iii)* The drift-metric battery's geometry pass needs the fine-tuned encoders themselves, which are excluded for size; its stored output is released, so the table is re-derivable even though the measurement behind it is not. **No third party has run any of this.** What we offer is re-derivation from committed run records and a bounded two-cell from-scratch check, not independent replication.

What the two-cell check found, and why we report it rather than a tolerance. `scripts/rerun_two_cells.sh` retrains one Moirai cell (Small/ETTh2 h=96, n=500, seed 42, full fine-tuning (B)) and one non-Moirai cell (TimesFM/ETTh1 h=24, n=1,000, seed 42, the same condition) from the raw CSVs and diffs every stored field. **TimesFM reproduces exactly**: all nine fields, including CKA and weight drift, are bit-identical. **Moirai reproduces its MSEs to within a few percent but not its forgetting percentage**: zero-shot MSE moves -0.8%, held-out MSE $+0.8\%$, the selection-window MSE $+3.2\%$, CKA -0.6% (0.957 to 0.952), and weight drift $+25.2\%$, while forg.$_{\text{B}}$ moves from $+6.81$ to $+11.14$, a 63.5% relative change. That last figure is the informative one and it is not a failure of the smaller ones, it is those smaller ones compounded: forg.$_{\text{B}}$ is $(\text{MSE}_{\text{val}} - \text{MSE}_{\text{ZS}})/\text{MSE}_{\text{ZS}}$ on two nearly equal quantities, so a -0.8% move in the denominator against a $+3.2\%$ move in the numerator's larger term is amplified into $+63.5\%$ in the ratio. This is a property of the percent-of-zero-shot scale rather than a fact about this cell: wherever the fine-tuned and zero-shot losses are close, a ratio of their difference to one of them magnifies absolute moves that are themselves small, which makes forg.$_{\text{B}}$ the least stable quantity this paper reports. **Δ_{enc} does not inherit that instability**: both arms share the same zero-shot denominator, so Δ_{enc} is the B-versus-D contrast rescaled per cell (Equation 1), and a shared positive rescaling cannot change its sign, which is what every claim in the paper is stated in terms of. Moirai scores the median of 20 sampled forecasts and TimesFM emits a point forecast, which is why only one of the two is reproducible to the bit. Every sign-bearing field kept its sign, which is what the paper's claims are stated in terms of, but a reader should treat any single-seed Moirai forgetting figure as carrying at least this much run-to-run variation, and that is why the paper reports cells at 3–10 seeds with a dispersion and never a single run.

Two things are required to reproduce the Moirai runs at all. First, the `uni2ts` `PackedStdScaler` performs an in-place write that silently detaches the gradient tape for losses depending on scaled statistics; without the patch of Appendix S1 training can appear to run while not converging. Second, the frozen-encoder condition is not a strict freeze (the input projections and mask encoding remain trainable, which is why its CKA is not 1.000), and the strictly-frozen variant is reported separately (Appendix A7). Comparisons against a "frozen" baseline are not comparable across papers unless which parameters were frozen is stated.

Scoring protocol. All headline comparisons are scored on windows disjoint from the ones used for selection. This is not a refinement: §5.3 shows that scoring on the selection windows instead reverses the sign of the frozen-encoder comparison in 4 of the 31 intervention cells, always in the same direction. Reproducing our numbers therefore requires reproducing the split discipline, and Appendix Table 8 gives the per-arm decomposition needed to check which split a given number was scored on.

Pre-registration. The prospective test of §4 was registered before any outcome run for those cells existed: `results/v47_prospective/preregistration.json` for the first batch and `results/v57_prospective3/preregistration_v3.json` for the second, each committed before the first condition-B or condition-D run of its own arm was launched. The scorer, `score_prospective.py`, reads those files and the finished runs and makes no decisions of its own. All are in the release, and each registration's commit precedes every run it scores. **That ordering is checkable in the version-control history, and the supplementary archive does not carry one.** What travels with the archive is each registration's own `written_utc` and the commit hash it was written at; the per-run records store no timestamp, so the archive alone cannot order them against it. We state the limit rather than let "checkable" cover a check the archive does not support: from the archive a reader has the hashes and our word, and the ordering itself is verifiable against the repository released with the paper. The Chronos/M4-Monthly re-run of Appendix S6 carries a second registration (`results/chronos_m4/preregistration.json`), on the same terms and with the same check available. That arm is *not* prospective (an earlier run on the cell had already been seen), so what its registration fixes is only what was still fixable: the gate's estimator and split, the seeds, the decision rule, and what each gate outcome would be taken to

mean. The gate was computed after that file was committed, which matters because the estimator was itself one of the things being corrected. The $N{=}4$ and $N{=}5$ rungs of Appendix A10 carry a third (`results/v49_layerunfreeze_10seed/preregistration_ladder.json`), committed before the first of their six runs. It is also not prospective, and what it fixes is the rungs, the seeds, the condition and flag, the reported statistic, and **what each outcome would mean for the body**: that a non-monotone forgetting profile across depth, or a thin rung landing outside the matched pair's range, would be reported as found with the body's claim narrowed to the matched pair rather than the ladder being dropped. Both of those outcomes occurred, and that is the rule we followed.

The figures check themselves. Both generated figures read their values from `scripts/cell_matrix.py` and *assert* at draw time every count the body quotes from them: the number of gate-passing cells, the number of degradation cells, the number of sign reversals and the sign of each correlation. A run record that moved would fail the figure rather than change it silently, which is the one staleness path the table emitters do not cover.

What we release. The analysis and experiment scripts, the per-run JSON records behind every table, the emitter for every table and the one-command re-derivation script, the benchmark-data manifest, every pre-registration file, and the LaTeX source and style files needed to build this paper. Model checkpoints are excluded for size; the appendices state which runs retained a checkpoint.

USE OF AI ASSISTANTS

An LLM-based coding assistant was used in this work: **a large fraction of commits in this project's version-control history carry an AI co-author trailer**.

Where it was used. Implementation and prose. It wrote the initial draft of most of the analysis and experiment scripts, including the table emitters, the one-command re-derivation sweep, the prose-number checker, the figure scripts, and the anonymity and bundle-inventory gates; and it drafted and edited some of the sections of the paper's text.

Where it was not used. It produced no experimental result. Every number in this paper comes from a run that executed and wrote a JSON record, and the registered checks in `scripts/check_paper_numbers.py` compare the prose against those records, never against anything a model asserted. The research question, the study design, the pre-registered predictions and decision rules, the choice of what to cut and what to report, and every claim the paper makes are ours.

Responsibility, and one caveat. We have reviewed all AI-assisted work in this paper, and we take full responsibility for everything in it, errors introduced during AI-assisted work included. Such errors did occur, and the two kinds we hit repeatedly are worth naming: tables whose numbers had been typed by hand and then drifted from the records they were supposed to summarise, and claims that were reworded in one place while a restatement elsewhere kept the old figure. Neither was caught by rereading the paper; each was caught by a mechanical check. That is the reason having a verification apparatus is critical, and it is the part of this methodology we would recommend to anyone writing a paper this way.

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APPENDIX

Notation. The appendices write the paper's estimand in its condition form, B−D, because they compare conditions directly (for instance B−D against B−H). It is the same quantity the body calls Δ_{enc}; the two are interchangeable throughout.

A1 WHAT THIS PAPER CLAIMS, AND WHAT IT DOES NOT

Three claims of unequal strength are made in this paper, and they are not interchangeable. **C1** is a null in *prediction*, established out of sample under an analysis registered before it was run. **C2** is a positive claim about what the screen does, at an operating point we chose rather than derived. **C3** is methodological, and it is the only one of the three we would defend outside these benchmarks.

A2 EXTENDED LIMITATIONS AND FUTURE WORK

(1) Single-backbone scope of the dissociation claim. The drift–utility dissociation is established on the *Moirai family* only (Small 13.8M, Base 91M, Large 311M across ETTh1, ETTh2, ETTm2). We attempted three further backbones as second-model controls across three ETT datasets (Table 2).

All nine backbone×dataset cells gate-fail (ZS under-performs the Linear baseline by 14% to 148%), consistent with ETT-family distributions being outside the pre-training coverage of these backbones at these sizes. The Chronos capacity sweep (Chronos-T5-Small/Base/Large on ETTh2) confirms the gap is size-invariant: scaling from Small (210M) to Large (710M) yields no improvement ($-42.6\% \rightarrow -34.3\% \rightarrow -35.2\%$). TimesFM-2.5-200M and MOMENT-1-base gate-fail across all tested datasets. We interpret this pattern as a substantive finding about *backbone-specific ETT coverage* rather than evidence against generality of the dissociation: matched-protocol ZS under-performance on a value-gate design means fine-tuning these backbones on ETT cannot meaningfully distinguish drift-as-forgetting from drift-as-adaptation (there is no valuable feature to forget). A gate-passing second backbone on a different dataset (Traffic, ILI, Exchange, or a model with explicit ETT pre-training coverage) remains the most direct path to a multi-backbone replication and is the highest-priority future-work direction.

(2) Sample-sweep depth and seed count. The sample-size sweep (Table 6) covers Moirai-Small/ETTh2, h=96, at $n \in \{500, 1\text{k}, 2\text{k}, 5\text{k}, 10\text{k}\}$. At n=10k the reported arm is the CUDA run: 10 seeds (42, 101, 123, 202, 303, 456, 777, 789, 888, 999), `-deterministic` (`CUBLAS_WORKSPACE_CONFIG=:4096:8`, `cudnn.deterministic=True`, seeded DataLoader, PYTHONHASHSEED), 10 epochs early-stopped at a mean best epoch of 4.2 ± 1.3. Three further n=10k records exist under other protocols, and Table 3 scores all four against the same held-out zero-shot reference. *Dispersions here are standard deviations over seeds (ddof=1), because the protocols differ in seed count.*

On held-out windows the four protocols agree: full fine-tuning at n=10k improves on the released checkpoint by 16–21%, in 27 of 28 seeds. On the selection windows the same records span -11.7% to $+10.6\%$ and change sign. That spread is the selection-window instability of §5.3, not a property of the stopping rule. A full n-sweep on Moirai-Base and Moirai-Large is future work.

(3) Out-of-domain retention is measured on the paper's benchmarks. The retention arm (§7.2) scores each checkpoint on the other four datasets of this study, not on Moirai's pre-training corpus, which is not released in a form we can evaluate on.

(4) Two pre-registered predictions were falsified. ILI's CKA prediction of 0.90–0.95 (Appendix A13), and the prospective value-gate predictor, which scored precision 0.00 at every threshold in $[0.10, 0.60]$ (§4). We report both as failures of our own diagnostic, which is what the pre-registration was for.

(5) Why similarity is not mechanism, in two directions. A representation can drift far and still carry the task information in a transformed basis, so a low CKA need not mean a capability was lost; and a small drift can destroy one critical direction, so a high CKA need not mean one was kept. Both directions are consistent with the positive control of Appendix A18, where CKA on task A's

Table 1: The paper's three claims with their evidence and their strength, and then the three readings we do *not* claim. The rows are ordered by how far each travels: C1 holds for the cells we evaluate, C2 for the screen at the threshold we state, C3 for any study that reads a post-treatment similarity statistic as evidence about a treatment's value. The three N rows are not hedges: each is a proposition this paper's own evidence is insufficient to support, and two of them would be supported by results we report if the scope restriction were dropped.

	Claim	Evidence	Strength and scope
C1	CKA magnitude does not identify whether encoder adaptation contributed, in the cells we evaluate.	within Moirai, the only arm with enough cells to ask, ρ=+0.168, clustered CI $[-0.469, +0.717]$, and no cut on CKA separates the cells encoder adaptation helped from the ones it hurt (§6). Out of sample, with the ladder fixed in advance, leave-one-cluster-out R^2 over the 15 clusters *falls* when CKA is added: it *lowers* R^2 from $+0.201$ to $+0.176$ (Appendix A17).	A null in *prediction*, not evidence of no relationship. The in-sample slope is $+43$ pp per unit CKA with a cluster interval of $[-31, +107]$, which admits a moderate positive relationship. Scope: the 31 intervention cells; the within-backbone form is Moirai only.
C2	The screen locates cells where no advantage over a fitted baseline is measurable under a definition we state.	only 7 of 31 cells show a pretrained advantage over a lookback-96 ridge *fit on the target data* (§4), and no cell beats every admissible rung of the eight-baseline ladder (Appendix A11).	Positive, but at a prespecified operating point of 0.20 we chose and did not derive, and V_b is a property of the model–baseline pair rather than of the checkpoint (§3). A cell the screen fails is one where *this* ladder finds no advantage.
C3	A post-treatment similarity statistic cannot be assumed to identify the value of a treatment that changed the representation.	Structural rather than statistical: the statistic is measured after the treatment and on the object the treatment altered (§3). Our results are consistent with it; they are not what establishes it.	The one claim we would defend outside these benchmarks. It licenses no threshold and predicts no effect size: it says a similarity statistic is the wrong instrument for this question whatever its value.
N1	*Not claimed:* that drift is generally unpredictive across time-series foundation models.	Three backbones cannot establish it, and pooled across backbones the same correlation is in fact clearly positive, which we report rather than set aside (§6).	Would require a backbone sample this paper does not have. C1 is a statement about our cells.
N2	*Not claimed:* that fine-tuning does not harm time-series foundation models.	No cell met a degradation definition fixed in advance, under any protocol we ran; that is a statement about these benchmarks and these protocols (Appendix S8).	A zero count at a stated definition, not a safety guarantee. Loosening the definition produces degradation cells, and we show which (Appendix S8).
N3	*Not claimed:* that mitigation methods are unnecessary.	LoRA, L2-SP and EWC were run as a full spectrum on one cell only, and we report what they did there (§5, Appendices A6, A8).	One cell does not generalise in either direction. Our claim is about the *diagnostic* that would trigger mitigation, not about mitigation.

inputs falls to 0.063 while task A itself *improves*. Our negative claim is about *identifying* damage from similarity alone, not about CKA's validity as a measure of similarity, which we do not dispute.

A3 Training Protocol per Trainer

The 31 intervention cells come from four trainers, and their stopping rules differ. Every entry is read from the stored run records except where marked ^{d}, which is the trainer's default, not overridden

Table 2: The initial non-Moirai screen, at h=96 with a lookback of 192, one seed (42) and 300 test windows. **This is a different protocol from the paper's non-Moirai screen** (Table 9), which runs h=24 at lookback 96 with three seeds and a paired frozen-encoder condition; the two are therefore not comparable cell by cell, and only the latter feeds the 32-cell count. Chronos-T5-Base/Large and MOMENT appear here and nowhere else. Advantage is V_{ridge} as a percentage. Rows trace to `results/v10_chronos_etth2.json`, `v14_*`, `v18_backbone_gate/*` and `v12_moment_etth2_gate.json`.

Backbone	Dataset	ZS MSE	Linear MSE	Advantage
Chronos-T5-Small	ETTh2	0.304	0.213	−42.6%
Chronos-T5-Small	ETTh1	0.118	0.092	−28.8%
Chronos-T5-Small	ETTm2	0.188	0.110	−70.2%
Chronos-T5-Base	ETTh2	0.286	0.213	−34.3%
Chronos-T5-Large	ETTh2	0.288	0.213	−35.2%
TimesFM-2.5-200M	ETTh2	0.242	0.213	−13.8%
TimesFM-2.5-200M	ETTh1	0.105	0.092	−13.9%
TimesFM-2.5-200M	ETTm2	0.190	0.110	−72.6%
MOMENT-1-base	ETTh2	0.528	0.213*	−148%

*MOMENT was screened at lookback 96, whose baseline is 0.235; its margin was nonetheless computed against the lookback-192 baseline 0.213, and we print the comparator actually used. Against 0.235 the margin is −125%, so the verdict is unchanged either way.

Table 3: Moirai-Small/ETTh2 h=96, n=10k, condition B, under four protocols. Held-out forgetting is against the held-out zero-shot MSE (0.4923); selection-window forgetting is the trainer's `forgetting_pct`. Mean±std over seeds.

Record	Protocol	k	Held-out forg.% (neg.)	Selection forg.%
`v19_cuda_etth2_n10k`	CUDA, 10 ep., early-stopped	10	−18.5±8.8 (10/10)	−5.3±6.6
`v18_mps_deterministic_n10k`	MPS, early-stopped	10	−21.3±6.3 (10/10)	−5.7±10.4
`v17_etth2_n10k_es`	MPS, early-stopped	5	−16.6±11.6 (4/5)	−11.7±5.3
`v16_etth2_n10k`	MPS, 20 ep., final epoch	3	−16.1±5.2 (3/3)	+10.6±9.8

by its launcher. Epoch selection, where it happens, uses the selection windows only; the held-out windows every outcome is scored on take no part in it.

Table 4: **Training protocol per trainer.** "Final" means the cell is read at its last epoch; "best" means the epoch with the lowest selection-window loss is restored before held-out scoring.

cells	trainer	n	lr	epochs	epoch read
Moirai, 21 cells	`finetune_forecasting.py`	500–1k	10^{-4}	20	final
Moirai-Small/ILI	`finetune_ili.py`	452	10^{-4}	20	best, patience 3
TimesFM-2.5, 4 cells	`finetune_timesfm.py`	1k	10^{-4}	20	best, patience 7^d
Chronos-T5, 5 cells, B	`chronos_mse_finetune.py`	8k	10^{-4}	30	best, patience 7^d
Chronos-T5, 5 cells, D	`chronos_frozen_converged.py`	8k	10^{-3}	≤3000	best, patience 200

Under D, Chronos trains only a head on cached frozen-encoder features, so it is run to convergence rather than for a fixed budget. The ILI cell also differs in checkpoint (Moirai 1.0-R-small) and lookback (104). The sample-size sweep (Table 6) is separate from these cells: its n≤5k points use the first row's protocol, and its n=10k point is early-stopped on a 10-epoch budget.

A4 Out-of-Domain Retention Arm

The design, thresholds and outcomes of this arm are fixed in `results/v59_ood_retention/preregistration.json`, committed before its first run. Checkpoints are scored with the positive control's evaluator (`scripts/eval_retention.py`), unchanged. PENDING: per-cell table and retention-versus-CKA figure

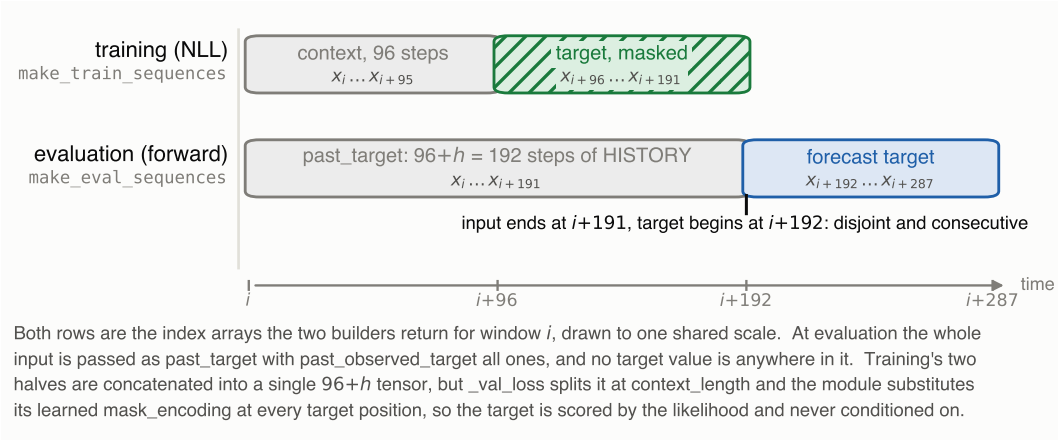


Figure 2: **Which timesteps Moirai is given, and which it is asked to predict**, at h=96; h=192 is the same layout with h substituted. Both rows are the index arrays that `make_train_sequences` and `make_eval_sequences` return for window i, drawn to one shared scale. At evaluation the model input is the $96{+}h$ steps *preceding* the forecast window and the target is the h steps strictly after it: the two are disjoint and consecutive, so no target value reaches the model. The figure is generated from those functions' output and asserts that disjointness at draw time.

A5 ETTh2 Forecasting Details

Dataset. ETTh2 (Electricity Transformer Temperature, Hourly, dataset 2) contains 17,420 timesteps of 7 features (HUFL, HULL, MUFL, MULL, LUFL, LULL, OT) recorded hourly from electrical transformers. We use the standard chronological split: 12 months training, 4 months validation, 4 months test.

Window construction, and exactly which values Moirai is given. The evaluation windows use an "extended lookback" of $96{+}h$ steps, which could be read as though the forecast target were part of the model input. It is not, and because the training and evaluation window sets differ the arithmetic is worth writing out. Let $x_1, \dots, x_T$ be the series and h the horizon. Both window sets are built by two functions in `scripts/finetune_forecasting.py`, named here so that this paragraph and the code can be checked against each other.

Training (`make_train_sequences`): window i has context $x_i, \dots, x_{i+95}$ and target $x_{i+96}, \dots, x_{i+95+h}$. The two are concatenated into one $96{+}h$ tensor and passed to Moirai's `_val_loss` with fixed patch size 32, which is the interface's required calling convention: `_val_loss` splits the tensor at `context_length` and marks the latter part with `prediction_mask`, and the module then substitutes its learned `mask_encoding` at every masked position and zeroes the corresponding observed flags. The target values are therefore scored by the likelihood and never conditioned on.

Evaluation (`make_eval_sequences`): window i has input $x_i, \dots, x_{i+95+h}$ (that is, $96{+}h$ steps of *history*), and target $x_{i+96+h}, \dots, x_{i+95+2h}$. The input's last index is $i+95+h$ and the target's first is $i+96+h$, so the two windows are disjoint and consecutive and no target value appears anywhere in the model input. Only the input is passed, as `past_target` with `past_observed_target` all ones; the point prediction is the median of 20 forecast samples with `patch_size='auto'`. "Extended lookback" thus means a longer *history*, not history-plus-target. Figure 2 draws both layouts from the index arrays those two functions actually return rather than from this description, and asserts the disjointness at draw time, so the figure cannot drift from the implementation.

The one asymmetry this does introduce is between models: Moirai is given $96{+}h$ steps of history where the fitted baseline of Appendix A12 is given 96. That direction favours the pre-trained model (more context lowers MSE_{ZS}, raises V_b and so admits *more* cells), and Appendix S7.3 measures it rather than arguing it, by giving the baseline the same $96{+}h$ history.

All data is fed to Moirai in raw scale (Moirai's internal PackedStdScaler handles instance normalization); predictions and targets are then normalized using training-set mean and standard deviation for MSE/MAE computation, matching the standard time-series forecasting evaluation protocol.

Condition C's objective and its contrastive labels. Condition B fine-tunes on NLL only; condition C adds a supervised contrastive term (Khosla et al., 2020), $\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda\, \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y}; \tau)$, with $\lambda = 0.5$ and $\tau = 0.07$. Condition C appears in this paper only as a protocol variant of B and carries no claim; the comparison every claim turns on is B against D. For the contrastive labels we divide training sequences into 8 temporal clusters based on their position in the time series. Sequences within the same cluster are positive pairs; sequences from different clusters are negatives. This encourages the encoder to maintain temporal-regime awareness.

Horizon selection. We evaluate h=96 and h=192, the two horizons at which Moirai-Small on ETTh2 clears the value gate: measured against the fitted lookback-96 baseline the gate is defined with (Appendix A12), V_{ridge}=+0.400 at h=96 and +0.464 at h=192 on held-out windows. Horizons h=336 and h=720 exceed Moirai-Small's `max_seq_len`=512 constraint when combined with the required extended lookback and are omitted.

A6 ETTh2 Drift Diagnosis and Mitigation (Full Tables)

This section contains the full tables referenced in §4. Table 10 shows the mitigation spectrum, conditions B through H on Moirai-Small/ETTh2 at both horizons.

Table 6 gives the sample-size sweep on ETTh2 (h=96, condition B) referenced in §6.

The full 31-cell intervention evidence base. Table 7 lists every cell carrying a paired B/D intervention behind the headline claim of §1: 31 of the 32 screened cells, the screen itself being a gate value and needing no intervention run: three backbones, six datasets, three Moirai capacities and two horizons, each with its selection-split gate and the intervention contrast B−D (positive = the frozen encoder was better). It is generated by `scripts/cell_matrix.py -latex` directly from the committed run JSON; no value in it is transcribed by hand. The two protocol arms (§4 at h=96/192, §6 at h=24) are listed together for completeness but are never pooled into a single statistic.

Why the four reverse in one direction, and the two mechanisms behind it. The four reversals of §5.3 do not share one per-arm story. Normalising each arm by the zero-shot error on its own split is an identity rather than a model, so the two arms' held-out-minus-selection moves account for the reversal exactly. On **two of the four** the frozen arm collapses out of sample and the selection-optimism reading holds: Chronos/ETTh1 is the largest reversal in the matrix at +6.8 on the selection windows against −39.2 held out, a 46.0-point swing (Δ frozen +53.9 against Δ adapted +8.0 on Chronos/ETTh1, the largest of the four). On the **other two** both arms finish *better* than their own zero-shot out of sample and nothing degrades at all — the adapted arm merely gains more — so half the reversals are a different phenomenon wearing the same sign change. What all four share is not a mechanism but a location: they concentrate where the two splits are least exchangeable: the *untrained* checkpoint's held-out-to-selection error ratio spans 0.98 to 5.07 across these four cells, a quantity that involves no fine-tuning at all. A frozen encoder cannot adapt to that shift and a fine-tuned one partly can, which is exactly the asymmetry that makes same-split scoring flatter freezing, so the direction of the four reversals is not a coincidence to be noted but a consequence to be expected, and it is why scoring Δ_{enc} on selection windows is not a defensible option rather than merely a less careful one.

The mitigation spectrum, cell by cell. Table 10 reports five strategies spanning unconstrained fine-tuning to full freezing, all on Moirai/ETTh2. The body (§6) states only the conclusion; the per-method readings are here. **LoRA** (r=8, α=16) improves over zero-shot at h=96 at CKA 0.979, but is not uniformly safe: on Moirai-Large it fails at the default learning rate (+25.7%) and only a 10× LR reduction rescues it (−8.5%, CKA 0.989), so tune the learning rate before the rank (Appendix S5). **EWC and L2-SP** reduce drift without buying utility: at n=1,000, h=96, each scored against its own zero-shot, they land at +13.4 to +16.0%, against the +11.1% full fine-tuning reaches at n=500. Those two figures are not matched on training-set size (no full-fine-tuning arm at n=1,000 on this cell has more than a single seed), so what we read off them is that anchoring buys no utility, not an ordering between anchoring and full fine-tuning. **Freezing** eliminates drift but cannot adapt,

Table 5: Baseline-relative pre-trained advantage $V_{\text{ridge}}=1-\text{MSE}_{\text{ZS}}/\text{MSE}_{\text{Linear}}$ for every Moirai cell, on both splits. The *selection* columns are the paper's primary gate: they are disjoint from the held-out windows every intervention outcome is scored on, so cell inclusion is not a judgement made after seeing the outcome. The *held-out* columns are the retrospective variant, reported as a sensitivity analysis. Bold marks a pass on the *primary* split in both columns of that row, so the bold set is the one the body's counts refer to. **Not a linear probe**: this is a ratio of forecast MSEs computed through Moirai's own forecasting head, and shares no machinery with a linear-probe ΔR^2 axis. **Estimator, exactly**: MSE_{ZS} is the median of 20 Moirai forecast samples at extended lookback (context + horizon); $\text{MSE}_{\text{Linear}}$ is a single lookback-96 ridge-OLS linear map $\mathbb{R}^{96\times D}\to\mathbb{R}^{h\times D}$ *fit on the cell's training windows* and applied unchanged out of sample; both are computed on the same matched windows within a split (300 selection, 300 held-out), train-normalised. The gate involves no fine-tuning, so a cell has one value at every n: the n=500 and n=10k (ES) operating points reported in §4 share the entry shown here. Bold = gate-pass at the 0.20 threshold on the primary split. "—" = not run.

		Selection (primary)		Held-out (retrosp.)	
Backbone	Dataset	h=96	h=192	h=96	h=192
Moirai-Small					
	ETTh1	−0.177	−0.347	−0.243	−0.331
	ETTh2	**+0.874**	**+0.932**	**+0.400**	**+0.464**
	ETTm2	−0.164	**+0.352**	−0.110	**+0.108**
	Weather	−0.546	−0.506	−0.867	−0.639
	`Electricity7`	−0.650	−0.455	−0.085	−0.047
Moirai-Base					
	ETTh1	−0.312	−0.600	−0.458	−0.501
	ETTh2	**+0.777**	**+0.925**	**+0.361**	**+0.275**
	ETTm2	−0.257	**+0.211**	−0.057	**+0.083**
	Weather	−0.823	−0.633	−1.029	−1.324
Moirai-Large					
	ETTh1	−0.248	—	−0.454	—
	ETTh2	**+0.875**	—	**+0.219**	—
	Weather	−0.638	—	−0.621	—

Reading: seven of these 21 cells clear the 0.20 threshold on the selection windows and five on the held-out windows. The selection set contains the held-out set and adds two (ETTm2, ETTm2), so the retrospective variant is the stricter of the two here. The primary survivors span two datasets: ETTh2, ETTm2. $V_{\text{ridge}}<0$ means the linear baseline wins outright. Appendix A12 compares these against an unfitted per-window trend baseline. `Electricity7` is Electricity restricted to 7 of its 370 series (six `MT` series plus `OT`), a deliberate protocol deviation that keeps the cell dimensionality-matched to ETT/Weather.

and the strict freeze of Appendix A7 shows that even the residual adaptation condition D permits is doing work. The screen is what makes this spectrum hard to act on: none of these methods has a demonstrated advantage to protect on 27 of our 32 cells, and on three of the remaining five the unconstrained condition beats both of the others run there: zero-shot by 31–42% and the frozen encoder by 20–22 points.

Condition coverage differs by cell, and we state it rather than let the widest row imply it. The full six-condition comparison exists only for Moirai-Small/ETTh2, so "best of six" is not a claim available for any other cell. Every intervention cell has B and D. Condition H (strict freeze) adds 16 cells (Appendix A7) and condition E (LoRA) adds 8 (Appendix A8), both on the seeds given in their own tables; L2-SP and EWC exist on Moirai-Small/ETTh2 alone. A claim about a cell is therefore a claim about the conditions that cell ran, and the per-cell seed counts are printed in every table rather than summarised into one figure.

A7 The Strict-Freeze Control

Condition D freezes the encoder's own weights but leaves the input projection (`in_proj`, 247,680 parameters) and `mask_encoding` (384) trainable alongside the distribution head (`param_proj`,

Table 6: **Sample-size sweep on Moirai-Small/ETTh2** (h=96, condition B). On held-out windows drift grows (CKA↓ as n ↑) while forgetting *falls*: more data moves the encoder further and leaves the model further ahead of zero-shot (10/10 seeds negative at n=10,000, Wilson CI $[0.72, 1.00]$). **If drift magnitude predicted harm the two columns would rise together; they move in opposite directions.** Scored on the selection windows instead (*sel.*), forgetting is non-monotonic in n, the selection-split effect of §5.3. Dispersion here is the sample standard deviation over seeds (ddof=1), not the standard error the rest of the paper reports, because the table is about how the spread across seeds moves with n.

Samples	k	held-out MSE	Forg.%	Forg.% (sel.)	CKA
500	3	.518±.096	+5.2±19.5	+14.0±6.7	.949±.010
1,000	3	.461±.019	−6.4±3.8	+13.4±9.1	.936±.025
2,000	3	.429±.019	−12.8±3.8	−2.9±5.9	.814±.042
5,000	7	.401±.015	−18.5±3.0	+6.6±10.0	.546±.082
10,000[cuda]	10	.401±.044	−18.5±8.8	−5.3±6.6	.518±.199

[cuda]The n=10,000 row is the CUDA arm (A10G): 10 deterministic seeds, 10 epochs *with* early stopping (mean best epoch 4.2±1.3). Every smaller n is MPS, 20 epochs *without* early stopping. That is the protocol difference Table 4 lists; Appendix A2 (2) gives the n=10,000 seeds and the matched MPS comparisons. Rows read `test_mse`, `forgetting_pct` and `final_cka` from the per-run records in `results/v8_final/n{500,1000,2000,5000}`, `results/v9_n5k_seeds` and `results/v19_cuda_etth2_n10k`.

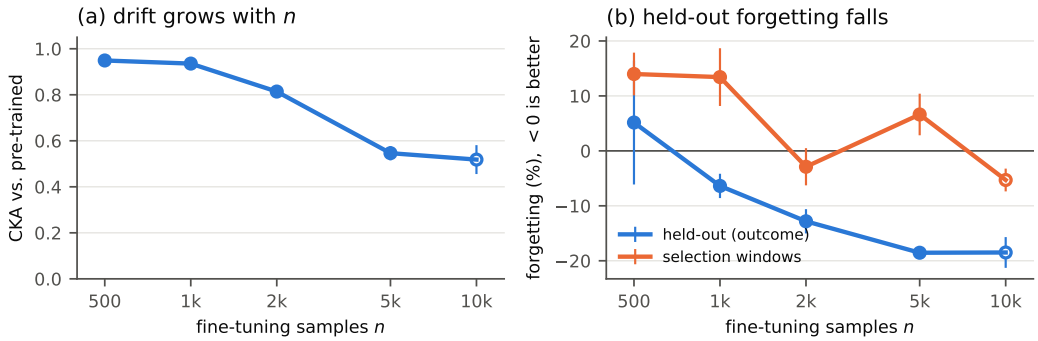


Figure 3: Sample-size sweep on Moirai-Small/ETTh2 (full fine-tuning, h=96), read from the same records as Table 6; error bars are standard errors over seeds. (a) CKA against the pre-trained encoder falls as n grows. (b) Held-out forgetting falls with it, so more drift goes with a better outcome; scored on the selection windows instead, forgetting is non-monotonic. Hollow markers at n=10k are the early-stopped arm.

2,956,800). What the encoder *receives* therefore keeps changing, so its output is not a fixed function of the input, which is why D's measured CKA is 0.76–0.90 on ILI and 0.97–0.99 on Moirai-Base rather than 1.000. A reader can reasonably object that wherever "freezing wins", it wins partly by re-fitting the input projection rather than by holding the encoder still.

Condition **H** removes the objection: `in_proj` and `mask_encoding` are frozen too, so only `param_proj` trains and the encoder's output is pinned by construction (measured CKA 1.0000 exactly, as it must be). B−H is then a clean read of what allowing *any* encoder-side change was worth, and B−D versus B−H separates encoder adaptation from input re-fitting. We fixed the reading rule before running: the claim is that the sign and reading of B−D survive strict freezing, so B−H is reported beside B−D on every cell, and divergence on any cell is reported as that cell's gap being partly input re-fitting rather than being buried.

Result (Table 11). The sixteen cells fall into three groups. Seven were chosen before the gate was corrected: they are the six that met the degradation definition under the unfitted trend baseline (Appendix A12), plus ILI. Nothing was dropped; two cells the prospective arm later added (Moirai-Large on ETTh1 and on Weather) postdate this control and carry no H run. That selection no longer

Table 7: All 31 intervention cells (the 31 of 32 screened cells with a paired B/D run): **selection-split** gate V_{ridge} (the primary gate of §4, scored on windows disjoint from the outcome columns beside it), then B−D on validation and on held-out windows (positive = the frozen encoder was better), and whether the validation and held-out readings disagree in sign. Every row names its own n because n is *not* constant down the column: the ETTh2 spectrum cells ran at 500 while their same-dataset neighbours ran at 1,000, and a row that stayed silent would read as sharing its neighbour's. ILI is the one cell with no cap: it trains on its whole 452-window split, since it is the only series small enough, so its n is a size rather than a subsample. Generated by `scripts/cell_matrix.py -latex`.

Cell	Seeds	Gate V_{ridge}	B−D validation	B−D held-out	Reverses?
Chronos / ETTh1 (h=24, n=8000)	3	-0.116 [f]	$+6.8\pm2.9$	$\mathbf{-39.2\pm13.9}$	yes
Moirai-S / ILI (h=24, n=452)	10	-10.206 [f]	-62.6 ± 1.4	$\mathbf{-38.6\pm1.3}$	no
Chronos / ETTm2 (h=24, n=8000)	3	-0.275 [f]	-24.1 ± 9.8	$\mathbf{-29.5\pm6.4}$	no
Moirai-B / ETTh2 (h=96, n=1000)	3	$+0.777$	$+3.8\pm4.1$	$\mathbf{-22.3\pm2.9}$	yes
Chronos / Weather (h=24, n=8000)	3	-0.203 [f]	-22.4 ± 4.5	$\mathbf{-21.6\pm3.2}$	no
Moirai-L / ETTh2 (h=96, n=500)	3	$+0.875$	$+7.2\pm5.8$	$\mathbf{-20.8\pm1.4}$	yes
Moirai-B / ETTh2 (h=192, n=1000)	3	$+0.925$	-3.9 ± 6.2	$\mathbf{-20.2\pm1.1}$	no
Chronos / ETTh2 (h=24, n=8000)	3	-0.184 [f]	-19.9 ± 3.2	$\mathbf{-12.9\pm0.6}$	no
TimesFM / ETTm2 (h=24, n=1000)	3	$+0.082$ [f]	-12.2 ± 12.4	$\mathbf{-7.7\pm2.6}$	no
TimesFM / ETTh2 (h=24, n=1000)	3	-0.208 [f]	$+0.2\pm2.9$	$\mathbf{-7.1\pm3.1}$	yes
Moirai-S / Electricity7 (h=96, n=1000)	3	-0.650 [f]	-10.4 ± 1.1	$\mathbf{-3.2\pm1.5}$	no
Chronos / Electricity (h=24, n=8000)	3	-0.089 [f]	-10.3 ± 3.2	$\mathbf{-2.7\pm1.9}$	no
Moirai-S / Electricity7 (h=192, n=1000)	3	-0.455 [f]	-8.2 ± 1.9	$\mathbf{-2.6\pm2.1}$	no
TimesFM / Weather (h=24, n=1000)	3	-0.160 [f]	-4.8 ± 0.6	$\mathbf{-1.6\pm1.5}$	no
TimesFM / ETTh1 (h=24, n=1000)	3	-0.101 [f]	-5.4 ± 0.9	$\mathbf{-1.4\pm2.8}$	no
Moirai-B / ETTm2 (h=192, n=1000)	3	$+0.211$	$+3.5\pm1.5$	$\mathbf{+1.3\pm3.4}$	no
Moirai-S / ETTm2 (h=96, n=1000)	5	-0.164 [f]	$+6.5\pm2.9$	$\mathbf{+2.6\pm3.4}$	no
Moirai-S / ETTh2 (h=96, n=500)	10	$+0.874$	$+10.6\pm2.8$	$\mathbf{+2.8\pm3.7}$	no
Moirai-S / ETTm2 (h=192, n=1000)	5	$+0.352$	$+8.6\pm3.3$	$\mathbf{+4.7\pm3.2}$	no
Moirai-S / ETTh2 (h=192, n=500)	10	$+0.932$	$+17.4\pm4.3$	$\mathbf{+6.1\pm3.4}$	no
Moirai-B / ETTm2 (h=96, n=1000)	3	-0.257 [f]	$+3.9\pm3.7$	$\mathbf{+7.0\pm2.4}$	no
Moirai-B / Weather (h=96, n=1000)	3	-0.823 [f]	$+19.3\pm0.8$	$\mathbf{+9.4\pm2.0}$	no
Moirai-L / Weather (h=96, n=1000)	3	-0.638 [f]	$+4.3\pm6.5$	$\mathbf{+12.0\pm7.5}$	no
Moirai-B / Weather (h=192, n=1000)	3	-0.633 [f]	$+30.6\pm12.0$	$\mathbf{+12.5\pm8.7}$	no
Moirai-S / ETTh1 (h=192, n=1000)	5	-0.347 [f]	$+14.8\pm1.9$	$\mathbf{+13.5\pm1.7}$	no
Moirai-S / ETTh1 (h=96, n=1000)	5	-0.177 [f]	$+16.1\pm1.5$	$\mathbf{+22.1\pm2.3}$	no
Moirai-L / ETTh1 (h=96, n=1000)	3	-0.248 [f]	$+13.4\pm2.1$	$\mathbf{+28.2\pm4.9}$	no
Moirai-B / ETTh1 (h=96, n=1000)	3	-0.312 [f]	$+15.8\pm4.6$	$\mathbf{+37.5\pm8.2}$	no
Moirai-B / ETTh1 (h=192, n=1000)	3	-0.600 [f]	$+15.6\pm2.3$	$\mathbf{+47.0\pm13.8}$	no
Moirai-S / Weather (h=96, n=1000)	3	-0.546 [f]	$+55.4\pm4.3$	$\mathbf{+53.1\pm10.2}$	no
Moirai-S / Weather (h=192, n=1000)	3	-0.506 [f]	$+70.2\pm12.3$	$\mathbf{+62.1\pm13.6}$	no

picks out a privileged class (against the fitted baseline no cell meets the definition), but it remains the set chosen to be *most* favourable to the degradation story, which is the useful property here. Five are the gate-passing survivors of §5.1, run afterwards precisely because the first seven did not reach them: until they existed, the survivors' readings rested on condition D alone. The last four are the Moirai/ETTm2 cells that clear the screen against a rung other than the fitted line (Appendix S10), added so that the control covers value-cells outside ETTh2: all four of them, at three seeds each, so ETTm2 is covered completely rather than partially. Measured CKA is 1.0000 everywhere, confirming the construction.

On twelve of the sixteen the reading survives; on four it reverses, and the four are the ones we added. B−H keeps B−D's sign on the original seven and on the five survivors. There the magnitude changes by -13.1 to $+28.9$ pp, and on the original seven by -5.8 to $+0.5$ pp, so at most 5.8 pp of a gap as large as 62 pp is input-projection re-fitting. On ILI, where adaptation helps, strict freezing makes adaptation look *more* valuable ($-38.6 \rightarrow -43.6$), which is the same direction: holding more of the model still costs more where the encoder was earning its keep. B−D is averaged over the seeds

Table 8: **The four sign reversals between selection and held-out windows** (§5.3). Δ adapted and Δ frozen are the change in each arm's forgetting when scoring moves from the selection split to the held-out split; zs_{test}/zs_{val} is the untrained checkpoint's error ratio between the two splits, an exchangeability check that involves no fine-tuning. All four reversals run from "frozen better" to "adapted better". n is per row and not constant (Table 7), and is taken from the published row rather than re-read here. Generated by `scripts/heldout_decomposition.py`.

Cell	B−D val	B−D held-out	Δ adapted (B)	Δ frozen (D)	zs_{test}/zs_{val}
Chronos / ETTh1 (h=24, n=8000)	+6.8	−39.2	+8.0	+53.9	0.98
Moirai-B / ETTh2 (h=96, n=1000)	+3.8	−22.3	+12.2	+38.3	2.31
Moirai-L / ETTh2 (h=96, n=500)	+7.2	−20.8	−52.9	−24.8	5.07
TimesFM / ETTh2 (h=24, n=1000)	+0.2	−7.1	−11.7	−4.3	1.09

Table 9: **Cross-backbone results (h=24)** (§6). Every non-Moirai cell with a paired frozen-encoder run: **selection-split** gate V_{ridge} ([f]=gate-fail at the 0.20 operating point; all nine fail, as does the screened-only TimesFM/Electricity cell, which has no paired run and so no selection-split gate: it scores +0.132 on the held-out windows), CKA after full fine-tuning, and the two outcome readings as % change against zero-shot (*negative is better than zero-shot*). **Protocol note:** this arm is univariate OT, lookback 96, h=24, 200 held-out windows, per-window z-scored; the Moirai arm of §4 is lookback 192, h=96, 300 windows, train-normalised. Gates for both arms come from one source (`results/gate_val_side.json`, via `scripts/cell_matrix.py`); the two arms are never pooled.

Backbone	Dataset	Gate V_{ridge}	CKA	Full FT	Frozen
Chronos-T5-S	ETTm2	−0.275 [f]	0.112	−41.4%	−11.9%
Chronos-T5-S	Weather	−0.203 [f]	0.092	−5.7%	+15.9%
Chronos-T5-S	Electricity	−0.089 [f]	0.232	−2.3%	+0.4%
Chronos-T5-S	ETTh2	−0.184 [f]	0.094	−0.4%	+12.5%
Chronos-T5-S	ETTh1	−0.116 [f]	0.169	+20.6%	+59.8%
TimesFM-2.5	ETTm2	+0.082 [f]	0.360	−51.8%	−44.1%
TimesFM-2.5	ETTh2	−0.208 [f]	0.246	−20.4%	−13.3%
TimesFM-2.5	Weather	−0.160 [f]	0.335	−7.8%	−6.1%
TimesFM-2.5	ETTh1	−0.101 [f]	0.561	−3.0%	−1.6%

condition H ran, so Moirai-Small/ETTh1's entries here are 3-seed values against the 5-seed values quoted in Table 7.

All four ETTm2 cells diverge, and we said in advance how that would be read. On every one of them Δ_{enc} changes sign between the two controls: Moirai-Small h=96 goes from B−D +7.3±2.9 to B−H −6.8±2.0, Moirai-Base h=96 from +7.0±2.4 to −9.2±3.1, Moirai-Small h=192 from +4.2±5.8 to −2.7±6.1 and Moirai-Base h=192 from +1.3±3.4 to −2.4±2.9: shifts of −14.2, −16.2, −6.9 and −3.7 pp. The last of those is the weakest of the four and we do not lean on it: both of its intervals straddle zero, so what that cell shows is a shift in the mean and not a resolved reversal. The pattern is carried by the two h=96 cells, where the shift is four to five times the SEM. The consistency of the *direction* across all four is what the reading rests on: four cells, one dataset, four shifts, all negative. The rule fixed before the runs was that divergence means part of that cell's gap is input re-fitting, and that is what these say: on ETTm2, condition D's apparent "freezing wins" was carried *entirely* by the input projection adapting to the target, and once it is held fixed too, allowing the encoder to adapt is the better choice. Two readings follow, and they point the same way as the rest of the paper rather than against it. First, the direction of the correction is towards adaptation on every diverging cell, so nothing here recovers a degradation case; ETTm2 joins ETTh2 in the column where full fine-tuning wins under the strict control. Second, and more important for §5, it is direct evidence that **condition D does not measure encoder adaptation alone**: on four of the sixteen cells where we can separate the two, the loose control's sign is set by the projections rather than by the

Table 10: **Mitigation spectrum on Moirai-Small/ETTh2.** Rows ordered by representation preservation (CKA at h=96). On *this* cell LoRA preserves representations (CKA$\approx$0.98) while *improving* on zero-shot, and weight anchoring reduces drift without buying utility. **Neither reading generalises, and we checked rather than assumed:** extending condition E to 8 further value-cells (Appendix A8) puts CKA_E anywhere from 0.55 to 0.97 and leaves full fine-tuning ahead of LoRA on 4 of the 8. **Every row is scored against its own zero-shot.** Forgetting% is the mean over seeds of each run's (fine-tuned − zero-shot)/zero-shot, so no row borrows another's denominator. MSE, CKA and ℓ_2 drift are means over seeds with population std (ddof=0); the body's SEM convention (§3) does not apply here. **The rows are not matched on training-set size.** Conditions B, C and D use n=500 training samples (zero-shot MSE .1256 at h=96, .1580 at h=192); E, F and G use n=1,000 (zero-shot MSE .1274 and .1596); the two rightmost columns give the seed count and the training-set size separately. We therefore claim no ordering *between* the two groups: there is no full-fine-tuning arm at n=1,000 on this cell with more than a single seed, so "anchoring ties full fine-tuning" is not a comparison these runs support. The dissociation the table is cited for (CKA not predicting utility) is within-group and unaffected. Condition D reads CKA 0.99996 rather than 1: it leaves `in_proj` and `mask_encoding` trainable, so it is a frozen-*encoder* control and not a strict freeze. Only condition H (Appendix A7) is 1.0000 by construction.

	$h = 96$				$h = 192$					
Method	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift	seeds	n
B: Full fine-tune	.140$_{\pm.014}$	+11.1	.937$_{\pm.028}$	3.96$_{\pm.56}$	.183$_{\pm.020}$	+15.7	.970$_{\pm.009}$	4.12$_{\pm.32}$	10	500
F: L2-SP (λ=0.01)	.150$_{\pm.006}$	+16.0	.939$_{\pm.023}$	4.98$_{\pm.34}$	.179$_{\pm.009}$	+12.7	.945$_{\pm.007}$	4.84$_{\pm.22}$	3	1,000
C: NLL+SupCon	.138$_{\pm.010}$	+10.1	.949$_{\pm.014}$	3.72$_{\pm.43}$	.178$_{\pm.011}$	+12.9	.964$_{\pm.015}$	3.97$_{\pm.34}$	10	500
F: L2-SP (λ=0.1)	.147$_{\pm.003}$	+13.4	.965$_{\pm.006}$	2.94$_{\pm.08}$	.181$_{\pm.003}$	+13.6	.980$_{\pm.002}$	2.84$_{\pm.02}$	3	1,000
G: EWC (λ=1000)	.149$_{\pm.015}$	+14.9	.972$_{\pm.004}$	3.29$_{\pm.79}$	.188$_{\pm.004}$	+18.3	.978$_{\pm.006}$	4.46$_{\pm.38}$	3	1,000
E: LoRA (r=8)†	**.116**$_{\pm.001}$	**−9.1**	.979$_{\pm.006}$	0.00$_{\pm.00}$	**.157**$_{\pm.002}$	**−1.8**	.986$_{\pm.005}$	0.00$_{\pm.00}$	5	1,000
D: Frozen encoder	.126$_{\pm.006}$	+0.5	.99996$_{\pm.00001}$	1.11$_{\pm.15}$	.155$_{\pm.002}$	−1.7	.99995$_{\pm.00002}$	1.30$_{\pm.18}$	10	500

† LoRA (α=16) adds low-rank updates that do not modify base encoder parameters; ℓ_2 drift is exactly zero by construction.

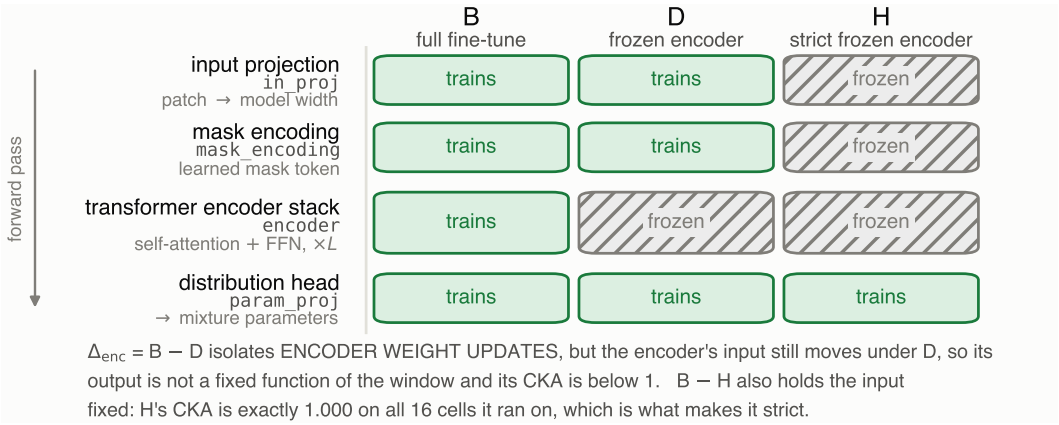


Figure 4: **Where the freeze boundary falls, module by module.** Green boxes are parameters the optimiser may move; hatched boxes are held at their pre-trained values. The grid is transcribed from the `requires_grad` assignments in `scripts/finetune_forecasting.py`. Two of our seven survivors change the sign of Δ_enc between D and H (Table 11), so which of these columns a paper means by "we froze the encoder" is not a detail.

encoder. That is a limit on B−D as an estimand, and it is why §3 defines Δ_enc against the control actually used and reports H beside it rather than substituting one for the other.

On the five survivors the control cuts against freezing, not for it. These are the cells the objection mattered most for, and on the three where full fine-tuning already won, strict freezing *widens* its margin rather than narrowing it: B−D of −22.3, −20.2 and −20.8 becomes B−H of −28.0, −31.8 and −33.9 on Moirai-Base h=96, Base h=192 and Large h=96, a move of −5.6 to −13.1 pp. Part of what condition D was scoring on these cells was therefore the input projection re-fitting

to the target data, and removing it makes encoder adaptation look *more* valuable, not less. On Moirai-Small h=192 (the cell that comes closest to degradation), B−H is $+1.2\pm8.5$ against B−D's $+4.1\pm8.6$: freezing still reads as better, by less, with a standard error several times the mean either way. Condition H also improves on the un-tuned model in every seed of all five (forg_H from -0.2% to -8.5%), so on the survivors both parts of the degradation definition behave the same way under the strict control as under the loose one.

The one large change is the diverged seed. Moirai-Small/ETTh2 h=96 moves from B−D $+13.7$ to B−H $+42.6\pm29.5$, the only cell whose magnitude shifts by more than 13 pp. It is seed 42, which §5.1 reports as a training failure rather than an encoder-drift effect: that seed diverges under full fine-tuning ($+98.9\%$) *and* under the frozen encoder ($+91.1\%$), but it cannot diverge under condition H, where only `param_proj` trains and the run therefore sits at zero-shot (-0.1%). Over the other two seeds B−H is $+14.4$ against B−D's $+16.7$. The strict control thus supplies independent support for the seed-42 diagnosis: what fails on that seed is optimisation, and holding the encoder's *inputs* fixed as well as its weights is what prevents it.

Table 11: The strict-freeze control (condition H) beside condition D on sixteen cells: the six that met the degradation definition under the unfitted trend baseline, plus ILI, plus the five gate-passing survivors of §5.1, plus all four Moirai/ETTm2 value-cells. **B−H keeps B−D's sign on twelve and reverses it on all four ETTm2 cells**, where condition D's margin was the input projection re-fitting rather than the encoder being held still. B−D is restricted to the seeds condition H ran, and a cell with fewer than three H seeds is excluded and named by the emitter rather than printed thin. n is shown on every row and is not constant down the column; ILI's 452 is its whole training split rather than a cap (Table 7). Generated by `scripts/cell_matrix.py -latex`.

Cell	Seeds	B−D	B−H	forg_H	CKA_H
Moirai-S / Weather (h=192, n=1000)	3	$+62.1\pm13.6$	$\mathbf{+56.7\pm13.5}$	-10.6 ± 0.3	1.0000
Moirai-S / Weather (h=96, n=1000)	3	$+53.1\pm10.2$	$\mathbf{+50.1\pm6.6}$	-12.8 ± 0.6	1.0000
Moirai-B / ETTh1 (h=192, n=1000)	3	$+47.0\pm13.8$	$\mathbf{+47.5\pm13.7}$	-8.0 ± 0.4	1.0000
Moirai-B / ETTh1 (h=96, n=1000)	3	$+37.5\pm8.2$	$\mathbf{+32.5\pm8.2}$	-9.7 ± 0.4	1.0000
Moirai-S / ETTh1 (h=96, n=1000)	3	$+23.2\pm3.6$	$\mathbf{+20.4\pm3.6}$	-0.6 ± 0.2	1.0000
Moirai-S / ETTh1 (h=192, n=1000)	3	$+14.5\pm1.4$	$\mathbf{+8.7\pm1.5}$	$+1.2\pm0.4$	1.0000
Moirai-S / ETTh2 (h=96, n=500)	3	$+13.7\pm9.0$	$\mathbf{+42.6\pm29.5}$	-0.2 ± 0.1	1.0000
Moirai-S / ETTm2 (h=96, n=1000)	3	$+7.3\pm2.9$	$\mathbf{-6.8\pm2.0}$	-2.9 ± 0.4	1.0000
Moirai-B / ETTm2 (h=96, n=1000)	3	$+7.0\pm2.4$	$\mathbf{-9.2\pm3.1}$	-3.9 ± 1.8	1.0000
Moirai-S / ETTm2 (h=192, n=1000)	3	$+4.2\pm5.8$	$\mathbf{-2.7\pm6.1}$	$+1.3\pm0.3$	1.0000
Moirai-S / ETTh2 (h=192, n=500)	3	$+4.1\pm8.6$	$\mathbf{+1.2\pm8.5}$	-1.4 ± 0.2	1.0000
Moirai-B / ETTm2 (h=192, n=1000)	3	$+1.3\pm3.4$	$\mathbf{-2.4\pm2.9}$	-6.0 ± 0.4	1.0000
Moirai-B / ETTh2 (h=192, n=1000)	3	-20.2 ± 1.1	$\mathbf{-31.8\pm1.0}$	-4.5 ± 2.0	1.0000
Moirai-L / ETTh2 (h=96, n=500)	3	-20.8 ± 1.4	$\mathbf{-33.9\pm0.1}$	-8.5 ± 0.5	1.0000
Moirai-B / ETTh2 (h=96, n=1000)	3	-22.3 ± 2.9	$\mathbf{-28.0\pm3.3}$	-3.5 ± 0.4	1.0000
Moirai-S / ILI (h=24, n=452)	10	-38.6 ± 1.3	$\mathbf{-43.6\pm0.7}$	-7.1 ± 0.5	1.0000

One clause does not survive, and it is not the headline one. "Degradation" has two parts: fine-tuning ends up worse than the un-tuned model, *and* freezing improves on it. The first part is a property of condition B and is untouched by this control. The second is condition D's, and on two cells the strictly-frozen model does *not* beat the un-tuned model: Moirai-Small/ETTh1 h=192 at $\text{forg}_\text{H} = +1.2\pm0.4\%$, positive in all 3 seeds where forg_D is -4.8%, and Moirai-Small/ETTm2 h=192 at $+1.3\pm0.3\%$, positive in all 3 seeds where forg_D is -5.8%. On those cells the head alone cannot recover the zero-shot forecast; D's improvement came partly from re-fitting the input projection. The other fourteen cells keep both parts (forg_H from -0.2% to -12.8%), so "freezing also beats not fine-tuning" holds strictly on 14 of 16. The first part is unaffected in either direction: where freezing wins it keeps winning under the strict control (B−H positive on all six candidate cells), and where full fine-tuning wins it wins by more (the five survivors above). We report the cells rather than re-defining the term around them.

A8 LoRA on the Value-Cells: the Constrained Middle of the Adaptation Axis

Conditions B and D bracket the adaptation axis: everything trains, or the encoder body does not. Neither tells a practitioner whether a *cheap constrained* adaptation would have beaten the extreme their cell happened to run. Condition **E** (LoRA, r=8, α=16) is that middle point. Beyond Moirai-Small/ETTh2 we ran E on the eight Moirai value-cells that lacked it, seeds $42/123/456$, each at its own arm's n and epoch budget (Table 12).

What E can and cannot measure, fixed before the runs. LoRA adds trainable low-rank updates to the encoder's attention projections, so the encoder's *function* changes while its base weights do not: ℓ_2 weight drift is exactly 0.0 on all 24 records by construction, and it is omitted from the table rather than printed as a column of zeros that would invite the reading that the encoder was held still. CKA is the measure that still reads on E. Δ_{enc} remains B$-$D; E is not a second estimate of it.

The one-cell reading does not survive. CKA_E spans 0.5482 to 0.9674 across these eight cells, against the $\approx$0.98 the single-cell spectrum reports, so LoRA is not a representation-preserving method in general; it is one on some cells. Full fine-tuning finishes ahead of LoRA (B$-$E < 0) on **4 of the 8**. LoRA is itself worse than not fine-tuning at all on two cells ($\text{forg}_\text{E} = +11.9\%$, 2 of 3 seeds, on Moirai-Base/ETTh2 h=96; $+11.2\%$, 3 of 3, on Moirai-Base/ETTm2 h=192).

Against the frozen encoder, read as a paired contrast. D$-$E is paired per seed rather than taken as the difference of two means, which matters: on Moirai-Small/ETTh1 the B$-$E and B$-$D means sit 0.2 pp apart with a SEM of 4.9, and a difference of means there would read as "LoRA beat freezing" on a cell that cannot separate them. Paired and requiring a unanimous sign, **LoRA beats freezing on 3 of 8 cells, freezing beats LoRA on 2, and 3 are inseparable**. Moirai-Base/ETTh2 h=192 is the one cell that orders all three points cleanly (full fine-tuning, then LoRA, then frozen), and it is a gate-passing cell, so the ordering is on a cell with something to preserve.

Why this belongs in a paper about drift as a diagnostic. It is a third condition on which the paper's central claim holds. Across these eight cells CKA_E does not order the outcome of constraining adaptation (ρ=-0.167 against B$-$E; -0.333 against B$-$D on the same cells): the highest-CKA cell here, Moirai-Large/ETTh2 at 0.9674, is one where the constrained option *loses* to full fine-tuning by 27.8 pp, and the lowest, Moirai-Base/ETTm2 h=96 at 0.5482, is one where it wins. As with B$-$D, how much the representation moved does not tell you whether constraining the move helped.

Two limits of this arm. The two Moirai-Small/ETTh2 cells already carrying E are excluded here: their E arm ran at n=1,000 against B/D at n=500, a mismatch disclosed in Table 10's caption and not improved by adding a third n. And α is not written into the run records (only r is), so α=16 is traceable to `scripts/run_lora_valuecells.sh` and its committed command line rather than to the per-run JSONs, the same limitation the learning rate carries.

A9 CKA Calibration: Random-Initialisation Baseline

A CKA of 0.09 is only interpretable against a floor: the value two *unrelated* encoders of this architecture score against each other. We measure it by re-initialising the Moirai-Small encoder randomly and comparing it to the pre-trained one over the same 200 ETTh2 validation windows and the same pooled-representation hook used for every other CKA in this paper (`scripts/cka_random_init_floor.py`, records in `results/v50_cka_floor/`). Restoring the pre-trained weights afterwards returns CKA 1.000000, so the comparison measures re-initialisation and nothing else.

The floor is not zero. The measured floor is **0.215**:

Why a naive re-initialisation reads zero is worth recording: "Xavier for weight tensors, zeros for biases" also zeroes LayerNorm's scale parameter, which annihilates the signal at the first normalisation and drives the encoder to a *constant* output. CKA is then 0 because one side has no variance, not because the two share no structure. Our script asserts that the randomised encoder's output variance is non-trivial, and reports 0.215 once normalisation parameters are left at γ=1; initialising every layer with its library default instead gives the same answer to within one standard deviation.

Table 12: **Condition E (LoRA, r=8, α=16) on the eight Moirai value-cells that lacked it.** All contrasts are held-out test MSE, normalised by the cell's zero-shot test MSE, in percentage points; $\pm$ is SEM over the seeds in the Seeds column. Sign convention matches B−D and B−H throughout: **positive means full fine-tuning did worse than the comparison arm**. B−D is recomputed on E's seeds, so it is comparable to B−E within a row and may differ from the same cell's row in Table 7, which averages that cell's full seed set. Parenthesised counts are the seeds sharing the mean's sign. ℓ_2 weight drift is exactly 0.0 for every run and is not shown: LoRA leaves the base parameters untouched by construction, which is why CKA, not drift, is the representation measure here. Rows ordered by D−E.

Cell	Seeds	B−D	B−E	D−E	forg_E	CKA_E
Moirai-S / ETTm2 (h=96, n=1000)	3	+7.3±2.9	+13.9±2.1	**+6.5±0.8** (3/3)	−23.6±0.3 (0/3)	0.8438
Moirai-S / ETTm2 (h=192, n=1000)	3	+4.2±5.8	+7.3±6.8	**+3.1±1.0** (3/3)	−8.6±0.9 (0/3)	0.9280
Moirai-B / ETTh2 (h=192, n=1000)	3	−20.2±1.1	−17.7±1.1	**+2.5±0.9** (3/3)	−18.6±3.0 (0/3)	0.6944
Moirai-S / ETTh1 (h=96, n=1000)	3	+23.2±3.6	+23.4±4.9	**+0.2±1.3** (2/3)	−3.6±1.1 (0/3)	0.9093
Moirai-B / ETTm2 (h=96, n=1000)	3	+7.0±2.4	+4.9±2.9	**−2.2±1.8** (1/3)	−18.0±2.7 (0/3)	0.5482
Moirai-L / ETTh2 (h=96, n=500)	3	−20.8±1.4	−27.8±0.7	**−7.0±1.2** (0/3)	−14.6±0.7 (0/3)	0.9674
Moirai-B / ETTm2 (h=192, n=1000)	3	+1.3±3.4	−19.6±7.0	**−20.9±5.3** (0/3)	+11.2±5.3 (3/3)	0.6495
Moirai-B / ETTh2 (h=96, n=1000)	3	−22.3±2.9	−43.3±14.1	**−21.0±11.4** (1/3)	+11.9±10.5 (2/3)	0.9173

Table 13: CKA calibration, measured. The floor is the CKA of a randomly re-initialised encoder against the pre-trained one (5 seeds each, mean±std, ddof=1); the two initialisation schemes agree. Reference rows are scoped to the cells they come from, since CKA at n=500 and at n=10k on the same cell differ by more than the floor itself.

Comparison	CKA	Source
Floor: random re-init, library-default	**0.215±0.054**	`v50_cka_floor`
Floor: random re-init, Xavier weights	**0.219±0.047**	`v50_cka_floor`
Moirai-S/ETTh2 h=96, cond. B, n=500 (10 seeds)	0.937±0.029	`ff_20ep`
Moirai-S/ETTh2 h=96, cond. B, n=10k (10 seeds)	0.465±0.123	`v49`
Chronos, cond. B, 4 cells at h=24	0.092–0.169	`v44`
Frozen encoder, cond. D, Moirai (134 runs)	0.889–0.9999	`v5-v57`
Strict freeze, cond. H (7/7 cells)	1.0000	`v45`

What this changes. "ETT fine-tuning preserves most pre-trained structure" cannot be read off small-n runs: the same cell at n=10k reads 0.465, and the range across all 31 cells is 0.092 to 0.970. The condition-D row also needs a scope: D is uniform only within Moirai, and its 134 runs exclude the partial-unfreeze arm of Appendix A10, whose runs are stored under the same `condition_D` filenames and reach down to CKA 0.317 while three to five of six layers train. Because D leaves `in_proj` and `mask_encoding` trainable (§3), TimesFM's D falls as low as 0.376, while Chronos's D and every condition-H run are exactly 1.0000; only H is 1.0000 by construction.

What it adds. Comparing the floor against all 31 cells, **four fall below it** outright (the Chronos cells at CKA 0.092, 0.094, 0.112 and 0.169), and two more (Chronos/Electricity 0.232, TimesFM/ETTh2 0.246) sit inside the floor's own seed-to-seed range (0.132 to 0.288). This is a measured basis for a claim §6 otherwise has to assert: on those cells CKA is at or beneath what an *untrained* encoder scores, so it cannot order them, and their outcomes span −39.2 to −2.7 in held-out B−D regardless. The remaining 25 cells, including all five gate-passing survivors and both arms of the layer-unfreeze comparison (0.716 and 0.465, Appendix A10), sit clear of the floor, so CKA differences among them are readable.

A10 Layer-Wise Unfreeze Localization

Moirai-Small has 6 transformer layers. The `-unfreeze-top-n-layers` flag of `finetune_forecasting.py` leaves the top N trainable, and we swept N on ETTh2 n=10k, h=96; N=0 is condition D and N=6 is condition B. N=3 and N=6 are 10 seeds each, and N=4 and N=5 are a pre-registered 3-seed top-up

(`results/v49_layerunfreeze_10seed/preregistration_ladder.json`), all in one root, protocol and environment: Apple MPS, torch 2.14, `PYTORCH_ENABLE_MPS_FALLBACK=1`, early-stopped with the best selection-window state restored. (Below torch 2.5, `uni2ts`'s declared pin, the NegativeBinomial component of Moirai's output mixture collapses to `total_count`$=0$ on MPS and training dies in the first NLL step.) $N{=}0$ is an older 5-seed record read at the final epoch, and is context rather than part of the comparison. At $N{=}1$ training diverged (NaN activations at epoch 11) at the default learning rate despite gradient clipping, and left no record.

Table 14: Layer-wise unfreeze ladder on Moirai-Small/ETTh2 ($h{=}96$, $n{=}10$k). N is the number of top transformer layers left trainable, so $N{=}0$ is condition D and $N{=}6$ is condition B. Held-out forgetting is against the held-out zero-shot MSE and is the outcome; selection-window forgetting is shown for contrast. Mean±SEM over seeds, with the count of seeds improving on zero-shot. $N{=}3$–6 share one root, protocol and environment and restore the best selection-window epoch; $N{=}0$ is an older record read at the final epoch. $N{=}1$ diverged (NaN at epoch 11) and is omitted.

N	k	CKA	ℓ_2 drift	Forg.% (held-out)	Forg.% (sel.)
0†	5	0.999±0.000	3.86±0.13	−5.3±0.2 (5/5)	−6.6±0.7 (5/5)
3	10	0.716±0.060	9.42±1.25	−9.5±2.6 (8/10)	−5.9±2.0 (9/10)
4	3	0.741±0.058	10.24±2.42	−16.5±3.1 (3/3)	−9.5±3.8 (3/3)
5	3	0.747±0.076	9.76±2.46	−17.7±0.6 (3/3)	−3.5±5.1 (2/3)
6	10	0.465±0.039	13.47±0.75	−22.9±2.0 (10/10)	−7.9±2.1 (8/10)

†Final epoch, no early stopping. The registered statistic for the $N{=}4, 5$ top-up is selection-window forgetting, mean±population std: $N{=}4$: −9.5±5.3; $N{=}5$: −3.5±7.3. The held-out column is a post-hoc reading. Rows read `test_mse`, `forgetting_pct`, `final_cka` and `final_weight_drift` from `results/v49_layerunfreeze_10seed` and `results/v18_cond_d_probe`.

Finding: on held-out windows, more unfreezing is better, and drift grows with it. Held-out forgetting falls with every rung, from -5.3% at $N{=}0$ to -22.9% at $N{=}6$ (10/10 seeds improving), while CKA falls from 0.999 to 0.465 and ℓ_2 weight drift rises from 3.9 to 13.5 (Table 14). The matched pair is separated by 13.3 points on a standard error of the difference of 3.3 (4.0 SE): unfreezing all six layers drifts further and forecasts better than unfreezing three. This is the sample-size sweep's pattern (Table 6) inside a single encoder, and the only claim we draw from the ladder. The two 3-seed rungs sit between the pair, in order, but are too imprecise to establish a shape. The held-out reading is post hoc: the top-up's registration fixed selection-window forgetting as its statistic, and on that statistic its registered non-monotone outcome fired, so the claim stays with the matched pair.

Scored on the selection windows the ladder has no order (-5.9, -9.5, -3.5, -7.9% across $N{=}3$–6) and the matched pair is indistinguishable. That is the selection-window instability of §5.3 again: the stopping rule picked each run's epoch on those windows, so their error is optimistic in a way that compresses differences between arms.

Per-layer CKA. CKA of each layer's mean-pooled output against the same layer of the untrained checkpoint, over the same 200 selection windows, on the $N{=}6$ set and on the CUDA set of Table 6 (`scripts/layerwise_cka_from_ckpt.py`, records in `results/v49_layerunfreeze_10seed/per_layer_cka_*.json`):

Layer	Re-run $N{=}6$ (v49)	Older CUDA set (v19)	Group
1 (bottom)	0.998±0.001	0.998±0.000	bottom 3
2	0.969±0.011	0.980±0.005	bottom 3
3	0.783±0.081	0.816±0.080	bottom 3
4	0.589±0.127	0.675±0.091	top 3
5	0.939±0.044	0.947±0.023	top 3
6 (top)	0.937±0.047	0.945±0.023	top 3
Bottom 3 mean	0.917	0.931	
Top 3 mean	0.821	0.855	

The drift is not monotone in depth: it concentrates in the middle of the stack, with a trough at layer 4 and near-identity at layer 1, and the top 3 layers drift more than the bottom 3. The layers the ladder adds at $N{=}4$, 5 and 6 are layers 3, 2 and 1, the *least* displaced under full unfreezing, so the CKA drop at $N{=}6$ cannot be charged to the layer that rung adds. Two scale caveats: computed token-flattened instead of mean-pooled, every layer reads ≥ 0.94 and the profile is too compressed to order; and the per-layer mean is not the global CKA of Table 14 (0.869 against 0.465), because the global figure compares the encoder's final pooled representation. We offer no mechanism; that would need per-layer gradient norms.

A11 Eight Denominators, One Numerator: Is the Screen a Property of the Estimator?

The screen's verdict barely moves when the 0.20 threshold is swept, but moves for 17 of 21 Moirai cells between the unfitted and fitted linear baselines (Appendix A12). The estimator is the parameter that matters, so this appendix varies it across a family, on *both* splits: the selection windows every admission decision in the paper is made on (Table 15, the 31 cells with a paired frozen-encoder run) and the held-out windows the retrospective variant scored (Table S8, all 32 screened cells). The body's counts are the selection-split ones.

What is held identical. $V_b{=}1{-}\mathrm{MSE}_{\mathrm{ZS}}/\mathrm{MSE}_{\mathrm{base}}$, and only the denominator depends on the estimator, so the numerator is read from the published gate for that split (`results/gate_val_side.json` or `gate_test_side.json`) and never recomputed: each column differs from the published one in exactly one respect. Before anything is reported, `scripts/gate_baseline_sensitivity.py` recomputes the `fitted` column through the alternative-baseline code path and checks it against both the published denominator *and* the published V_b; all 32 cells agree to a relative tolerance of 10^{-6}, and the script refuses to emit a table if any does not. The two checks are not redundant, and the ILI cell is why: its denominator matched while its V_b did not, because that cell aggregates as a mean over features of per-feature ratios where every other cell is a ratio of means (`gate_all_cells.ili_gate`, matching `finetune_ili`). A denominator check alone would have published one row of this table on a convention no other number in the paper uses.

The estimators. *Persistence* repeats the last observed value for all h steps and is the bottom of the ladder. *Seasonal naive* repeats the last seasonal period and fits nothing; periods are declared per dataset (24 steps for the hourly ETT, Weather and Electricity series, 96 for 15-minute ETTm2, 52 for weekly ILI) rather than inferred, because inferring one from an autocorrelation peak returns 1 on the flatter series and quietly substitutes a persistence baseline. *Per-feature AR* is the paper's map with the cross-channel terms removed, so it isolates how much of the gate rests on multivariate information. *DLinear* is a single linear map on the trend/seasonal decomposition of the lookback, shared across channels. *MLP* is one two-layer network per channel (hidden 128) with the penalty selected per channel on a chronological holdout; the channels that hit the 200-iteration budget are recorded per cell. *Tuned ridge* is the paper's own estimator with λ selected on a chronological 20% holdout of the training windows rather than fixed at 10^{-4}; the selected values span 10^{-6} to 10^{4}, and on all five ETTh2 cells the selection lands at 10^{3} or 10^{4}, so the fixed 10^{-4} the paper uses is badly under-regularised there. *Boosted trees* is one `HistGradientBoostingRegressor` per feature over that feature's lags plus the horizon step as an input feature; one model per (feature, step) would be 672 fits on a 7-feature $h{=}96$ cell. Its rows are windows$\times$steps, capped at $200{,}000$ per feature, which on the Moirai cells uses 1,041–2,083 of the 10,165–48,777 available windows. Every cap and every selected penalty is recorded per cell in `results/gate_baselines.json` rather than only in the source.

Four columns coincide on the single-channel arms, and that is arithmetic rather than a copy. On the Chronos and TimesFM cells, which score one target channel, the fitted map, per-feature AR and DLinear are the *same estimator*: with $F{=}1$ there are no cross-channel terms for AR to drop, and DLinear's trend/seasonal split is an invertible linear reparameterisation of the lookback, so a shared linear head on it spans exactly the maps a lookback regression spans. Their columns therefore agree to three decimals by construction, and tuned ridge differs from them only through λ. A reader who sees four identical entries on those rows is seeing the ladder collapse where the data has one dimension, not a transcription error.

A11.1 THE LADDER ON THE SELECTION WINDOWS (THE PAPER'S PRIMARY)

Table 15: **Gate V_b under eight denominators and a floor, on the *selection* windows.** Same conventions as Table S8; this is the split every admission decision in the body is made on, and the numerator comes from `results/gate_val_side.json`. Generated by `scripts/gate_baseline_sensitivity.py -split val`.

Cell	Fitted ridge (ours)	Training mean	Persist-ence	Seasonal naive	Per-feature AR	DLinear (shared)	Tuned ridge	MLP (128)	Boosted trees
Moirai-S / ETTh2 (h=192)	**+0.932**	**+0.943**	**+0.267**	+0.081	**+0.734**	**+0.331**	**+0.929**	**+0.889**	+0.971†
Moirai-B / ETTh2 (h=192)	**+0.925**	**+0.937**	+0.192	−0.013	**+0.706**	**+0.263**	**+0.922**	**+0.878**	+0.968†
Moirai-L / ETTh2 (h=96)	**+0.875**	**+0.953**	**+0.316**	+0.134	**+0.677**	**+0.243**	**+0.877**	**+0.897**	**+0.953**
Moirai-S / ETTh2 (h=96)	**+0.874**	**+0.953**	**+0.308**	+0.124	**+0.673**	**+0.234**	**+0.876**	**+0.896**	**+0.952**
Moirai-B / ETTh2 (h=96)	**+0.777**	**+0.917**	−0.220	−0.545	**+0.423**	−0.350	**+0.781**	**+0.817**	**+0.916**
Moirai-S / ETTm2 (h=192)	**+0.352**	**+0.716**	+0.092	+0.128	**+0.228**	+0.092	**+0.351**	**+0.497**	+0.723†
Moirai-B / ETTm2 (h=192)	**+0.211**	**+0.654**	−0.105	−0.062	+0.061	−0.106	**+0.210**	**+0.387**	+0.663†
TimesFM-2.5 / ETTm2 (h=24)	+0.082	**+0.217**	+0.041	+0.263†	+0.082	+0.082	−0.029	+0.412†	+0.598†
Chronos-T5-S / Electricity (h=24)	−0.089	+0.161	+0.100	+0.295†	−0.089	−0.089	−0.089	+0.102	−0.069
TimesFM-2.5 / ETTh1 (h=24)	−0.101	**+0.343**	+0.154	**+0.294**	−0.101	−0.101	−0.109	−0.035	−0.050
Chronos-T5-S / ETTh1 (h=24)	−0.116	**+0.358**	+0.157	**+0.322**	−0.116	−0.116	−0.124	−0.006	−0.039
TimesFM-2.5 / Weather (h=24)	−0.160	**+0.489**	**+0.437**	**+0.241**	−0.160	−0.160	−0.161	−0.071	+0.014
Moirai-S / ETTm2 (h=96)	−0.164	**+0.766**	+0.069	−0.252	−0.372	−0.394	−0.175	**+0.614**	+0.794†
Moirai-S / ETTh1 (h=96)	−0.177	**+0.385**	**+0.356**	+0.050	−0.225	−0.176	−0.198	−0.189	+0.501†
Chronos-T5-S / ETTh2 (h=24)	−0.184	**+0.461**	+0.488†	**+0.223**	−0.184	−0.184	−0.153	−0.091	−0.044
Chronos-T5-S / Weather (h=24)	−0.203	**+0.454**	**+0.430**	+0.165	−0.203	−0.203	−0.204	−0.145	−0.058
TimesFM-2.5 / ETTh2 (h=24)	−0.208	**+0.489**	+0.492†	**+0.213**	−0.208	−0.208	−0.182	−0.121	−0.051
Moirai-L / ETTh1 (h=96)	−0.251	**+0.347**	**+0.315**	−0.009	−0.302	−0.250	−0.273	−0.264	+0.470†
Moirai-B / ETTm2 (h=96)	−0.257	**+0.747**	−0.006	−0.352	−0.482	−0.505	−0.269	**+0.583**	+0.778†
Chronos-T5-S / ETTm2 (h=24)	−0.275	**+0.477**	+0.140	+0.527†	−0.275	−0.275	−0.293	−0.019	−0.295
Moirai-B / ETTh1 (h=96)	−0.312	**+0.315**	**+0.282**	−0.058	−0.365	−0.311	−0.335	−0.325	+0.444†
Moirai-S / ETTh1 (h=192)	−0.347	+0.100	+0.214†	−0.028	−0.443	−0.369	−0.347	−0.344	+0.633†
Moirai-S / Electricity7 (h=192)	−0.457	**+0.785**	+0.797†	−0.144	−0.809	−0.748	−0.485	−0.832	+0.868†
Moirai-S / Weather (h=192)	−0.506	**+0.242**	+0.339†	+0.121	−0.340	−0.233	−0.489	−0.284	+0.743†
Moirai-S / Weather (h=96)	−0.546	**+0.239**	+0.363†	+0.127	−0.436	−0.409	−0.537	−0.207	**+0.234**
Moirai-B / ETTh1 (h=192)	−0.600	−0.070	+0.066†	−0.222	−0.714	−0.627	−0.601	−0.597	+0.564†
Moirai-B / Weather (h=192)	−0.632	+0.178	+0.284†	+0.048	−0.453	−0.337	−0.614	−0.392	+0.721†
Moirai-L / Weather (h=96)	−0.639	+0.193	+0.325†	+0.074	−0.523	−0.494	−0.629	−0.279	+0.188
Moirai-S / Electricity7 (h=96)	−0.651	**+0.810**	+0.814†	−0.142	−0.789	−0.732	−0.669	−1.004	**+0.797**
Moirai-B / Weather (h=96)	−0.819	+0.104	+0.250†	−0.027	−0.690	−0.658	−0.808	−0.420	+0.099
Moirai-S / ILI (h=24)	−10.206	−0.518	+0.015†	−12.093	−10.206	−10.206	−11.072	−9.025	−6.171
Clearing 0.20	7/31	24/31	18/31	8/31	6/31	4/31	7/31	10/31	20/31
and admissible	7/31	24/31	8/31	5/31	6/31	4/31	7/31	9/31	5/31
Below the floor	0/31	0/31	12/31	3/31	0/31	0/31	0/31	1/31	15/31
Degradation cells	0/31	5/31	3/31	0/31	0/31	0/31	0/31	0/31	1/31

The fitted map is admissible everywhere; the two rungs that are not are the cheap ones. On the selection windows the fitted map clears the training-mean floor on all 31 cells, per-feature AR, DLinear and tuned ridge on all 31, and the MLP on 30. Persistence falls below the floor on 12 cells, boosted trees on 15 and seasonal naive on 3; for the trees the reason is the same as on the held-out windows, that they cannot extrapolate past their training range. Counting only admissible passes, the eight rungs give **7, 8, 5, 6, 4, 7, 9 and 5 of 31** on this split.

Read as a graded axis, the union spans all three backbones. With V_{best} measured against the strongest admissible rival each cell faces (a minimum over rungs), **no cell of the 31 clears** 0.20 **against every admissible rung**, and **20 clear it against at least one: 14 Moirai, 3 Chronos, 3 TimesFM**. That union is what §6 uses as its analysis scope, and it matters for a specific objection: the seven cells that clear the fitted ridge are all Moirai on two series, so on the fitted rung alone the gate-positive evidence looks confined to one backbone. It is not: 13 of the 20 union cells *fail* the fitted ridge and are admitted by some other admissible rung, including every Chronos and TimesFM cell in the union. The seven that do clear the fitted ridge have V_{best} between -0.545 and $+0.134$, five of them set by seasonal naive and two by DLinear.

On this split the definition does have positive instances, under two weak rungs. With clause (i) taken from each estimator and clauses (ii) and (iii) unchanged, the count is zero under the fitted

map, seasonal naive, per-feature AR, DLinear, tuned ridge and the MLP; **three under persistence** (Moirai-Small/ETTh1 h=96 at n=1k, Moirai-Base/ETTh1 h=96, Moirai-Large/ETTh1 h=96, with forg.$_{\text{B}}$=+18.5, +22.8 and +14.1 unanimously and forg.$_{\text{D}}$=−3.5, −14.7 and −14.1 unanimously); and **one under boosted trees** (Moirai-Small/Weather h=96, forg.$_{\text{B}}$=+37.3 in 3/3, forg.$_{\text{D}}$=−15.8 in 3/3). We report these rather than absorbing them. Three facts bound how much they carry. All four are ETTh1 or Weather cells, the same datasets as the eight instances the unfitted baseline produces (Appendix A12). Both admitting denominators sit just above the floor: persistence scores 1.053 against the training mean's 1.104 on ETTh1, and the trees 0.5363 against 0.5397 on Weather, so each is within 5% of a constant and roughly twice the fitted ridge's error (0.576 and 0.266) on the same windows. And every one of the four has *no* demonstrated advantage against a rung that is not near-constant: the fitted ridge puts them at −0.18 to −0.55 and the strongest admissible rung each faces at −0.23 to −0.55. The direction of the sensitivity is therefore the same as between the unfitted and fitted baselines: what manufactures a degradation cell is a *weaker* baseline, not a stronger one revealing damage we missed. Taking that trend to its limit confirms it from the other side: under the training-mean floor itself the count is **five**, the four above plus Moirai-Small/Weather h=192, and all five are among the eight the unfitted baseline admits.

A12 Fitted Versus Unfitted Linear Baseline

Two estimators that share a name. The capability screen is $V_{\text{ridge}} = 1 - \text{MSE}_{\text{ZS}}/\text{MSE}_{\text{ridge}}$ with threshold 0.20, where $\text{MSE}_{\text{ridge}}$ is a lookback-96 linear map *fit on the cell's training windows*: a single ridge-OLS map from $96{\times}D$ inputs to $h{\times}D$ outputs, applied unchanged out of sample (`gate_linear_baseline.fit_linear_map`). A common alternative that is also called a "linear baseline" fits a slope and intercept to the last 96 steps *inside each evaluation window* and extrapolates them forward (`linear_forecast`), using no training data. This appendix shows why the choice matters: it is the first of the two measurement prerequisites in §3.

The unfitted estimator is weaker than a constant. Table 16 measures four denominators on the same 300 matched test windows, on the train-normalised scale of the stored zero-shot MSE. On ETTh1 h=96, predicting the training mean scores 0.9937 against the trend extrapolation's 1.2207. A baseline that a constant beats cannot support the inference the screen exists to license, that there is a pre-trained capability here worth preserving. The fitted estimator is stronger by a factor of about three on ETTh1 and 9.4 on Weather. Its ETTh1 h=96 MSE of 0.4145 is in line with published multivariate Linear/DLinear results (Zeng et al., 2023). The screen keeps one asymmetry: Moirai sees lookback + h steps of context while the linear map sees 96, so the screen is a lower bound on the linear model's strength (Appendix S7.3).

Table 16: Baseline strength: train-normalised MSE on 300 matched test windows. Row 1 is the unfitted per-window estimator; row 4 is the denominator §3 defines. Row 1 is beaten by predicting the training mean (row 3) on *both* cells and is several times weaker than row 4. On the train-normalised scale the constant-mean predictor's MSE is just the mean squared normalised target, so row 3 needs no fitting; a value near 1.0 is what an on-distribution test split should give.

Denominator estimator	ETTh1 h=96	Weather h=192
Per-window trend extrapolation (`linear_forecast`)	1.2207	5.0707
Persistence (last observed value held)	1.1984	1.2227
Constant training mean	0.9937	0.8619
Fitted lookback-96 regression (`fit_linear_map`)	**0.4145**	**0.5403**

What the choice moves, cell by cell (`gate_all_cells.py -baseline {fitted,trend}`, held-out windows):

- **Moirai: 5 of 21** cells clear 0.20 under the fitted baseline, against 20 of 21 under the unfitted one. Seventeen cells change status: 16 flip pass→fail, and one (Moirai-Large/ETTh2 h=96) flips fail→pass at +0.219.
- **Chronos: 0 of 5**, against 4 of 5: ETTh1 −0.123, ETTh2 −0.325, Weather −0.103, ETTm2 −1.281, Electricity +0.048.

- **TimesFM: 0 of 5**, against 5 of 5: ETTh1 -0.101, ETTh2 -0.287, Weather -0.077, ETTm2 -0.945, Electricity $+0.132$.
- **ILI** (Moirai-Small): $\mathbf{-1.625}$ (Appendix A13).

Across arms, **7 of 31 cells pass on the selection split (the paper's primary), and 5 of 32 on the held-out variant**. Every survivor under either is Moirai; the selection split adds two ETTm2 cells to the five ETTh2 ones. Per-cell Moirai values are in Table 5, and the held-out survivors are in Table 17. Four of the seven survivors *improve* on the released checkpoint under full fine-tuning *in every seed*, three of them by 31–42% and the ETTm2/Base cell by 8.4%. The two whose mean is harmful have only 4/10 and 5/10 seeds in that direction. The seventh, Moirai-Small/ETTm2 $h{=}192$, is the mirror image: its mean is a 1.10% *improvement* but 3 of its 5 seeds are harmful. We count unanimity rather than means throughout, because a cell whose sign depends on which seeds were drawn supports no statement in either direction.

The $h{=}96$ cell's mean is one seed. Seed 42 of Moirai-Small/ETTh2 $h{=}96$ diverges to $+98.9\%$ under full fine-tuning *and* to $+91.1\%$ under the frozen encoder, so it is an optimisation failure rather than an encoder-drift effect; under the strict freeze it cannot diverge at all (Appendix A7). Dropping it leaves $\text{forg.}_{\text{B}}{=}-0.20{\pm}4.15$ over 9 seeds (3 positive) and $\text{forg.}_{\text{D}}{=}-2.45{\pm}0.16$. We nonetheless report every cell mean including seed 42, because a rule that discards runs after seeing their outcome is not one we are willing to apply.

Table 17: The five cells that still clear $V_{\text{ridge}}{\geq}0.20$ on held-out windows under the fitted baseline. Forg.% is percent change against zero-shot, so a *negative* entry means fine-tuning improved on the released checkpoint. The bracketed counts are the seeds whose sign matches the degradation direction: positive for B, negative for D. B$-$D positive means the frozen encoder was better.

Cell	Gate	B Forg.% (+)	D Forg.% ($-$)	B$-$D val	B$-$D held-out
Small/ETTh2 $h{=}192$, $n{=}500$	$+0.464$	$+1.99$ (5/10)	-4.16 (10/10)	$+17.4$	$+6.1$
Small/ETTh2 $h{=}96$, $n{=}500$	$+0.400$	$+9.71$ (4/10)	$+6.91$ (9/10)	$+10.6$	$+2.8$
Base/ETTh2 $h{=}96$, $n{=}1000$	$+0.361$	-31.44 (0/3)	-9.11 (3/3)	$+3.8$	-22.3
Base/ETTh2 $h{=}192$, $n{=}1000$	$+0.275$	-36.35 (0/3)	-16.12 (3/3)	-3.9	-20.2
Large/ETTh2 $h{=}96$, $n{=}500$	$+0.219$	-42.42 (0/3)	-21.62 (3/3)	$+7.2$	-20.8

No cell meets the degradation definition. The definition (gate ≥ 0.20, *and* condition-B forgetting positive in every seed, *and* condition-D forgetting negative in every seed) returns zero cells under the fitted baseline. There is one candidate, and the unanimity rule excludes it:

```
excluded:  Moirai-small/ETTh2 h192 n500   gate=+0.464
   forg_B=+1.99±3.32SEM (5/10 pos)
   forg_D=−4.16±0.20SEM (10/10 neg)   B-D=+6.15
=> 1 excluded for seed disagreement
```

Half its seeds go the other way on condition B. Appendix S8 shows that weakening the rule, in seven directions and at both gate thresholds, does not recover a cell either. Under the unfitted baseline the same definition returns eight cells, all on ETTh1 or Weather at $h{\in}\{96, 192\}$, and all eight fail the screen once it is fitted: the measurement choice alone produces the degradation instances.

Prospective arm and threshold sweep. With no positive instances, the pre-registered prospective test has nothing for the gate to find: on the eight pre-registered cells it scores **TP 0, FP 8, precision 0.00**, and recall is undefined. The threshold sweep (Table S2) gives precision 0.00 at *every* threshold in $\{0.10, 0.20, 0.30, 0.40, 0.50, 0.60\}$, with $8/8/6/5/3/2$ cells flagged, so the failure is not a badly chosen operating point.

Within Moirai ($n{=}22$ cells), the gate correlates with held-out B$-$D at Spearman $\rho = -0.518$: higher gate goes with adapting the encoder having been the better choice. The interval excludes zero when cells are resampled, $[-0.840, -0.040]$, and includes it when the six (backbone, dataset) clusters are, $[-0.810, +0.500]$; we report the latter and read only the direction (Appendix A16). The CKA comparison is gate-independent: $\rho = +0.168$, clustered CI $[-0.469, +0.717]$ within Moirai.

Reproducing either screen. `gate_all_cells.py` takes `-baseline {fitted,trend}` with `fitted` as the default, and the unfitted val-

ues are kept in `results/gate_test_side_trend.json` and `results/gate_val_side_trend.json`. The evaluation windows are built with `ext_lb` = lookback + h, matching the extended lookback the fine-tuning script evaluates with.

A13 THE ILI CELL

The cell is Moirai-1.0-R-Small on national influenza-like illness: 966 weekly observations, 7 features, h=24, lookback=104, split 579/193/194 chronologically, 10 seeds, early-stopped with patience 3 (Table 4). At ≈450 training windows it is a case study rather than a pillar.

ILI fails the screen by a wide margin. Against the fitted baseline the held-out gate is $\mathbf{-1.625}$, the value in Table 7. On the selection split the mean is -10.206, but that mean is dominated by NUM. OF PROVIDERS, whose near-straight validation segment the ridge map fits to MSE 0.034 and which alone contributes -44.0. The median per-feature reading is -3.40 on selection windows and -1.51 held out, and **0 of 7 features clear** 0.20 **on either split**, so the verdict is robust where the mean is not. ILI is therefore not a degradation cell, and no claim in §6 rests on its gate.

Adaptation helps, and most of the gain needs the encoder. Under full fine-tuning the encoder restructures heavily (CKA 0.456 ± 0.008) and the forecast improves by $-80.0\pm0.6\%$ on selection windows and $-50.77\pm0.49\%$ held out (10/10 negative on both). With the encoder frozen (condition D, the 3.2M parameters outside it trained), the forecast improves by $-12.18\pm1.28\%$ held out (10/10 negative), and the gap B−D is -62.6 ± 1.4 on selection windows and -38.6 ± 1.3 held out (10/10 negative on both). D reaches CKA 0.76–0.90, not 1.000, because the projections feeding the encoder remain trainable (Appendix A7). Table 18 gives the per-seed values.

Table 18: ILI per-seed results (Moirai-Small, h=24, lookback=104), both conditions.

	B (full fine-tuning)		D (frozen encoder)	
Seed	CKA	Forg.% (val / held-out)	CKA	Forg.% (val / held-out)
42	0.458	−78.6 / −51.5	0.804	−16.2 / −14.6
101	0.465	−81.4 / −50.1	0.898	−13.5 / −6.7
123	0.476	−80.0 / −48.9	0.756	−27.6 / −19.5
202	0.457	−81.4 / −52.1	0.791	−19.7 / −13.6
303	0.409	−78.1 / −49.0	0.823	−18.3 / −11.5
456	0.463	−75.5 / −50.0	0.851	−15.4 / −13.2
777	0.437	−81.6 / −52.5	0.822	−18.0 / −12.4
789	0.449	−80.3 / −49.5	0.865	−12.7 / −4.9
888	0.505	−82.0 / −50.8	0.848	−16.1 / −12.8
999	0.437	−80.8 / −53.4	0.807	−16.1 / −12.7
Mean	0.456	−80.0 / −50.8	0.827	−17.4 / −12.2

A falsified prediction. The pre-registered CKA prediction for this cell, 0.90–0.95 (intermediate restructuring), is **falsified**: CKA is 0.456. What it establishes is narrow: screen strength does not linearly predict drift magnitude. ILI's low CKA is consistent with its small training set, which permits aggressive restructuring within the stopping budget.

A14 PAIRED INFERENCE ON ENCODER ADAPTATION, AND WHAT THESE SEED COUNTS CAN RESOLVE

A unanimity count is not inference. This appendix reports Δ_{enc} per cell as a paired estimate with an interval, an effect size, two p-values and a multiplicity-adjusted q; states the three-way call the body is built on; and then says plainly what the matrix's seed counts can and cannot resolve.

The call, and why it has three outcomes. Thirty-one cells are tested, so we adjust: Benjamini–Hochberg across the cells within this contrast gives the q column, and a cell is called decisive only at $q<0.05$. That yields **2 cells where freezing is decisively better, 4 where adapting is decisively better, and 25 inconclusive**. The third category is the substantive addition. Our published criterion

recorded a cell with no measurable effect and a cell with a large effect its seeds disagreed about identically, as "not degrading"; under the interval rule the second says *inconclusive*, which is what it is, and the minimum detectable effect below is what separates the two.

Where the decisive cells sit relative to the screen. Both cells where freezing is decisively better, Moirai-Small/ETTh1 at h=96 (Δ_{enc}=+22.1, q=0.011) and h=192 (+13.5, q=0.014), **fail** the screen, at V_{ridge} of -0.177 and -0.347. Two of the four cells where adapting is decisively better, Moirai-Base/ETTh2 h=192 (-20.2, q=0.018) and Moirai-Large/ETTh2 h=96 (-20.8, q=0.022), **pass** it. So on this matrix the screen admits cells where adaptation decisively wins and rejects the only cells where preservation decisively wins. That is a sharper statement against the screen than the unanimity tally could make, and it is stated here because it counts against our own instrument. It also has exactly one exception at the augmented seed counts of Table S13, reported there rather than here because every number in this paragraph is at the published n.

Both degradation criteria are reported, and both return zero. Clause (i) of the definition in §3 is unchanged. Replacing the unanimity clauses (ii) and (iii) with interval clauses (the CI on Δ_{FT}=B−ZS entirely above zero, the CI on Δ_{frz}=D−ZS entirely below) selects **zero cells**, exactly as the pre-registered unanimity version does. The two rules can disagree in principle, in either direction, so the agreement is a result and not a tautology; `scripts/paired_inference.py` computes both and cross-checks the unanimity list against `cell_matrix.degradation_cells`, which is the definition of record, rather than trusting two spellings of one rule to stay in step.

The zero-shot reference's own error is propagated, and only where it belongs. On the 21 Moirai cells the zero-shot denominator is a per-cell mean over condition-A seeds rather than each seed's own measurement, so the B−ZS and D−ZS intervals above are widened by that mean's SEM, $\text{se}=\sqrt{\text{se}^2_{\text{within}} + \text{se}^2_{\text{ZS}}}$. Δ_{enc} is *not* widened and must not be: the reference cancels from its numerator and survives only as a common scale factor. That asymmetry is one of the reasons Δ_{enc} is the estimand. The widening is material on 13 cells and is nominally zero on 8, where the reference ran a single condition-A seed; on those the reference's error is *unestimated*, not zero, and we say so rather than let a 0.00 read as a check that passed.

What these seed counts could have detected. Table 20 turns "the interval is wide" into a number for the seven cells the screen admits, which are the only cells about which we make a preservation recommendation. At the multiplicity-adjusted level the call actually uses, **4 of the 7 are unresolvable within our feasibility cap of 20 paired seeds**: they would need 29, 36, 182 and 212. Two of the remaining three are already decisive. So on the cells where a practitioner would consult this screen, the honest summary is that our design settles two of seven and cannot settle four of seven at any budget we could run, which is a limitation of this matrix, not evidence that the effects are absent, and it is the reason §3 reports an interval rather than a sign count.

The exact test's resolution is set by N alone. A sign-flip permutation test on N paired differences has 2^N assignments, so its smallest attainable two-sided p is $2/2^N$: 0.25 at N=3, 0.0625 at N=5, 0.002 at N=10. The paired-N histogram across the 31 cells is {3:24, 5:4, 10:3}, so **only 3 of 31 cells can reach p<0.05 without a distributional assumption**, and only one does (ILI, p=0.002). Thirteen cells have a t-interval excluding zero where the exact test cannot reject at any effect size. We report both columns rather than the more favourable one: the t-test's rejections are real inferences under its normality assumption, and the exact column is the price of not making that assumption.

Why the reported interval is the t-interval, not the bootstrap. At N=3 every percentile-bootstrap resample mean lies inside $[\min, \max]$ of the three differences, so a same-sign triple excludes zero *by construction*. Such an interval reports the sign of the data, not the uncertainty of the mean. The bootstrap is computed and stored in `results/paired_inference.json`; the table prints the paired t-interval beside the exact p.

The aggregate straddles zero. Averaging cell means and resampling (backbone, dataset) clusters, the 20 value-cells give Δ_{enc}=+0.7 with CI $[-11.1, +14.1]$ and 10 of 20 positive; all 31 give +2.9 with CI $[-12.7, +15.1]$ and 16 of 31 positive. Neither excludes zero in either direction. This is the honest aggregate statement of the paper's central contrast: across these benchmarks we cannot establish that allowing encoder adaptation systematically helps *or* hurts, which is a different and weaker claim than either regime law the literature offers.

A15 Cross-Cell Statistics, and a Pooled Analysis We Do Not License

Across the 31 intervention cells of Table 7 we ask whether either observational diagnostic (CKA, or ℓ_2 weight drift) orders the held-out intervention contrast B−D. Two predictors rather than one is deliberate: if *both* fail, the statement is about observational measures of fine-tuning in general and not about CKA specifically. The statistic is Spearman ρ with a 10,000-resample bootstrap CI (seed 0), reported at two resampling levels: over cells, and over the (backbone, dataset) clusters that Appendix A16 argues is the correct level, and emitted by `scripts/clustered_inference.py`. Pooled over all 31 cells:

Predictor	n	Clusters	ρ	95% CI over cells	95% CI over clusters
CKA	31	15	$+0.567$	$[+0.265, +0.759]$ excludes 0	$[+0.239, +0.828]$ excludes 0
ℓ_2 weight drift	25	9	$+0.118$	$[-0.300, +0.518]$ includes 0	$[-0.489, +0.545]$ includes 0

The ℓ_2 row covers 25 of the 31 cells because the five Chronos cells and ILI retain no weight-drift measurement in their run records. No p-values are reported in this appendix or the next: cell-level resampling mis-states the dependence between cells, and this analysis is descriptive (Appendix A16).

The pooled CKA row is not a licensed analysis, and it is the row that would help us. The three backbones occupy almost disjoint CKA ranges (Moirai 0.36–0.97, Chronos 0.09–0.23, TimesFM 0.25–0.56), *and* differ systematically in outcome: every Chronos and TimesFM cell favours adaptation, and every large positive B−D is Moirai's. A rank correlation pooled over 31 cells drawn from three such groups therefore measures which backbone a cell came from at least as much as how far its encoder moved: it is a textbook setting for a group-level artefact of the Simpson's-paradox kind, and CKA magnitude is representation-dependent, which is why this paper compares CKA only within a backbone (§6). We stress the direction of the incentive. The pooled figure is the *favourable* one (it is the only reading in this appendix under which a diagnostic we introduced orders the intervention with an interval excluding zero), and the body nevertheless leads with the within-backbone figure. We decline to make the positive claim the pooled number would support, because the restriction that forbids it was fixed before these cells were run rather than chosen after seeing them.

Within Moirai the interval is wide, and we say so rather than round it to zero. On the 22 Moirai cells $\rho{=}{+}0.168$ with a cell-level CI $[-0.328, +0.598]$, widening to $[-0.469, +0.717]$ over the six clusters, a width of 0.93 on a scale whose whole range is 2, and 1.19 once the dependence is propagated, which is the honest bound this evidence supports. That interval comfortably admits a moderate positive relationship, and 22 cells cannot rule one out. The claim we license is the one the body makes: *CKA does not order our cells*. It is not that CKA is uninformative in general, and a reader who expects a real within-backbone relationship to exist will find nothing here that contradicts them; what they will not find is a cell ordering they could have acted on.

The sharpest form of the dissociation, inside one capacity. Restricting further to Moirai-Base (eight cells, one backbone, one capacity, so neither architecture nor parameter count varies), the ordering CKA gives is not merely uninformative but *reversed* (Table 22). Sorted by CKA, condition-B forgetting runs monotonically from -36.3% at the most drifted cell to $+39.5\%$ and $+22.8\%$ at the two least drifted, and those two least drifted cells are the only two in the arm that degrade at all. A practitioner reading "CKA is low, the encoder has been restructured, intervene" would have intervened on exactly the six cells that did not need it and left alone the two that did. Eight cells cannot establish the reverse relationship either (this is the same interval as above, restricted), which is the point: the sign of the association is not stable enough within a single backbone and capacity to carry a decision.

Chronos and TimesFM cannot be asked at all. With $n{=}5$ and $n{=}4$ the bootstrap resamples only a handful of distinct rank permutations and the interval spans nearly the entire admissible range, $[-1.000, +1.000]$ and $[-0.778, +1.000]$. Both rows are tabulated for completeness and neither carries information. In particular the $+0.400$ on four TimesFM cells must not be read as agreement with the pooled sign: it is one of very few values the statistic can take on four cells, and its interval covers almost everything.

ℓ_2 weight drift does no better, and it is the second of the two diagnostics we defined. Pooled (already the reading most likely to manufacture a correlation), weight drift gives $\rho{=}{+}0.118$ with a CI that includes zero at both resampling levels, widening from $[-0.300, +0.518]$ over cells to $[-0.489, +0.545]$ over its nine clusters. Since §3 defines exactly these two representation diagnostics, their joint failure is the more useful statement: it is not that we picked the wrong summary of the encoder's movement, but that neither summary we defined orders the outcome. The two are also not redundant and can move independently: a stack whose weight drift is *exactly* zero can read CKA well below one when the projections feeding it are trainable, which is what condition D does on TimesFM, where ℓ_2 drift over the transformer stack is 0.00 by construction while its measured CKA runs 0.38–0.89 (§3, §6; the analogous Moirai control is Appendix A7). A geometric battery (RBF CKA, SVCCA, orthogonal Procrustes, mean cosine and effective-rank change, on the 59 encoders that retained checkpoints) is Appendix S11. None of those measures orders the outcome either, and they do not order the *encoders* alike, so the failure is not a property of the summary we happened to pick.

A backbone fixed-effects reading, and its limits. The natural alternative to subsetting is to absorb backbone identity and let CKA enter as a continuous regressor, which answers the objection that we adopted the split that removes the significant relationship. We do fit it (Table 23, `scripts/cka_fixed_effects.py`): the within-backbone slope of held-out B$-$D on CKA is $+39.1$ pp per unit CKA with backbone fixed effects and $+39.9$ once CKA$\times$backbone interactions are added, against $+47.7$ with no backbone term at all. Both adjusted intervals include zero once clusters rather than cells are resampled. So the reduction from $+47.7$ to $+39.1$ shows that absorbing backbone identity does *not* by itself destroy the pooled slope (most of it survives adjustment), while the interval shows that 31 cells drawn from 15 clusters cannot pin the surviving slope down. What we do *not* do is fit variance components: with 22, 5 and 4 cells in the three groups there is no repeated-measures structure to estimate them from, so a mixed model over three groups of that size is not identified and any interval it produced would be an artefact of the specification. Cells here are unequally powered and not independent, so every cross-cell number in this appendix is descriptive.

A16 Cells Are Not Independent: Clustered Intervals for Every Cross-Cell Statistic

Resampling *cells* would treat each cell as an independent observation. They are not. A cell is a (backbone, size, dataset, horizon) combination, so the 31 cells draw on 3 backbones and 7 series: the Moirai-Small, -Base and -Large runs on ETTh2 at $h{=}96$ and $h{=}192$ share the series, the chronological split, the normalisation constants and (at fixed size) the pretrained checkpoint. Resampling cells therefore counts shared facts repeatedly and produces intervals that are too narrow, and a p-value computed from that scheme is not conventional inferential evidence. This is a fair objection to how the paper reported these numbers and it is answerable without new runs.

The resampling unit. We take the cluster to be the $(\text{backbone}, \text{dataset})$ pair: the level at which the pretraining corpus, the series, the split and the checkpoint family are shared. Clusters are drawn with replacement and all of a drawn cluster's cells enter together (`scripts/cluster_keys.py`, one definition shared by `cell_matrix.py` and `cka_fixed_effects.py` so the two cannot drift apart). A draw spanning fewer than three distinct predictor values admits no rank correlation and is discarded, which is why the draws kept are reported alongside every interval; on the 31-cell and 22-cell analyses the discard rate is under 0.1%, and it is material only on the four- and five-cell arms, which carry no information anyway. The effective sample size falls sharply: 31 cells are 15 clusters, the 22 Moirai cells are 6, and the 25 cells with a weight-drift record are 9.

What changes, in full. Of the eleven cross-cell correlations reported in this paper, **exactly one changes verdict** when the interval is taken over clusters instead of cells, and it changes against us. Both levels appear side by side in Table 21 and in the two tables of Appendices A15 and S12; the summary is:

- **Unchanged, and these carry the body's claims.** CKA does not order B$-$D within Moirai at either level ($\rho{=}{+}0.168$; cells $[-0.328, +0.598]$, clusters $[-0.469, +0.717]$). ℓ_2 weight drift does not order it at either level, pooled ($\rho{=}{+}0.118$) or within Moirai ($\rho{=}{+}0.318$). forg_B still orders it with an interval far from zero ($\rho{=}{+}0.894$; clusters $[+0.594, +0.965]$),

which is expected, since that relationship is close to an identity (Appendix S12). The unlicensed pooled CKA correlation still excludes zero, and we still decline to read it.

- **Changed: the gate's anti-predictive interval.** On the retrospective test-side gate, $\rho{=}-0.518$ between the gate and B−D moves from $[-0.840, -0.040]$ over cells to $[-0.810, +0.500]$ over the six Moirai clusters, so we report only the direction, as a description of these 22 cells. On the primary selection-split gate the correlation is $\rho{=}-0.293$ and no level excludes zero. The mechanism (the highest-gate cells are the ETTh2 and ETTm2 cells where adaptation pays best) remains visible cell by cell in Table 7 and does not rest on the interval.

Backbone fixed effects. The cluster bootstrap also lets us report the adjustment the previous version of this appendix declined to fit (Table 23).

Most of the unadjusted slope survives adjustment ($+47.7$ falls only to $+39.1$), so it is not the case that absorbing backbone identity destroys the pooled relationship, and we do not claim it. What the intervals say is that 15 clusters cannot pin the surviving slope down. **A wide interval is not evidence of absence.** At this cluster count an interval including zero means the design cannot resolve the sign, which is weaker than "CKA carries no information" and is the only thing we assert.

What this design could have resolved. "The design cannot resolve the sign" is a claim with a size, so we report the size. *Post hoc* — this analysis was specified after seeing the interval it explains, and is not part of the pre-registration — we simulate this exact design and ask for the smallest true Spearman correlation at which *our own* clustered interval excludes zero in 80% of realisations: $\mathrm{MDE}_{80} \gtrsim 0.5$. The observed $\rho{=}+0.168$ is less than half of it. Only one decimal is quoted: four master seeds place the figure between 0.5 and 0.6 and do not agree on a second digit, which `--seedcheck` reports rather than leaving to be discovered. So the reading this appendix asserts is **un-usability at this sample size, not absence**: six clusters could not have resolved a correlation of the size CKA appears to have, and a reader who believes CKA's true association with Δ_{enc} is 0.3 should conclude that we could not have seen it, which is a different sentence from ours and is compatible with it.

And more cells of the same kind would not fix it. On normal scores the intracluster correlation of CKA is 0.05 while that of Δ_{enc} is 0.74: within Moirai, CKA varies almost entirely *within* cluster (across model size and horizon) while the outcome varies almost entirely *between* clusters, across series. **A predictor with no cluster-level signal cannot order an outcome that is almost all cluster-level**, so the width of the interval is a property of where the two quantities' variance sits and not only of n — and adding Moirai sizes or horizons on the series we already have would add predictor variance in the direction that carries no outcome signal. This is also the second, independent reason this paper reads CKA within a backbone rather than pooled (§6, Appendix A15).

How the number was obtained, and the three ways it cuts against us. The observed CKA is conditioned on rather than simulated, so the real cluster sizes, spread and tie structure are used; only the outcome's latent is drawn, as $rz + \sqrt{1-r^2}\,(\sqrt{t}\,C_{\text{cluster}} + \sqrt{1-t}\,W_{\text{cell}})$ with z the normal scores of the observed CKA ranks. Because Spearman is rank-invariant **no marginal for Δ_{enc} is assumed at all**, and the assumption reduces to a monotone association with a Gaussian residual carrying an exchangeable cluster effect; r is calibrated to ρ by simulation rather than by the bivariate-normal inverse, which is measurably wrong on a fixed 22-point score vector. Three things then work against our own claim and are reported rather than buried. First, the headline residual cluster fraction is the *estimated* one, and it gives **the smallest MDE of the three settings we ran** at every one of the four master seeds, so the figure we quote is the least favourable point of the sensitivity range for our own claim — which is the only answer we can give to "you tuned the simulation" that is checkable rather than asserted, and the emitter asserts it. Second, every analytic cross-check is *higher*: a Fisher-z minimum detectable correlation with the Bonett–Wright factor gives 0.58 at $n{=}22$, 0.60 at the Moulton effective n of 20.1 and 0.93 at the six clusters, and inverting the observed half-width gives 0.59. The simulation is authoritative where they disagree, because it simulates the rule that produced our verdict and they describe a test we never ran — but the direction of the disagreement is a measured defect in our own estimator, and we report it: the six-cluster percentile interval is mildly anti-conservative, rejecting a true zero in 7% of realisations against a nominal 5%, and 10–11% in the two sensitivity settings; these are Monte-Carlo estimates on 300 and 100 realisations and reseeding moves them by about a point, so what they establish is that the interval is mildly anti-conservative and not by how much. **So the figure we quote is the optimistic end**, and a correctly sized rule would

put the MDE *higher* and this paragraph's claim on firmer ground rather than weaker. Third, an MDE bounds what a *rank* statistic could have seen, and no rank statistic detects a non-monotone relation at any n; the evidence bearing on that case is the pre-specified out-of-sample ladder of Appendix A17 and the decision pricing of Appendix S13, neither of which is rank-based. We do not report observed power at the observed ρ: conditioning power on the estimate it is meant to judge is circular, and the emitter records it under `not_a_claim` for that reason.

Two things this appendix does not do. It does not fit variance components: with 22, 5 and 4 cells in the three backbone groups a mixed model is not identified, and any interval it produced would be a property of the specification. And it does not recompute p-values at the cluster level. A p-value would need a null distribution for a rank correlation among dependent observations at an effective sample size near 6, and the paper describes this analysis as descriptive; printing $p{=}0.0136$ beside that sentence was a contradiction, so the p-columns are removed rather than re-derived.

A17 Does CKA Add Predictive Information Beyond Backbone Identity? A Pre-Specified Out-of-Sample Test

A wide interval on a within-backbone slope shows that a design cannot resolve a sign; it does not show that the predictor is uninformative, and the previous appendix says so in those words. That leaves the reader with a genuine question we had not answered: *if you already know which backbone you are on, the horizon and the training-set size, does measuring CKA improve your prediction of the value of encoder adaptation?* This appendix answers the decision-relevant version of that question out of sample.

The analysis was fixed before it was run, and the guarantee is weaker than the prospective arm's. The model ladder, the covariate construction, the deciding statistic and the sentence this paper would print under each of three possible outcomes were all written to `results/preregister_loco.json` and committed before any of the models was fitted. That is a real guarantee against specification search, and it is *not* the guarantee of §4: there the outcome runs did not yet exist, whereas here the outcome (held-out B−D) was already computed and already published in Table 7. What is frozen is the *analysis*, not a prediction about unseen data, and the registration file says so on its first page. The registered prediction was $\Delta R^2 \leq 0$.

The ladder, and why dataset identity is absent from it. Table 24 fits five nested models to Δ_{enc} over all 31 scored cells, validated by leaving out one (backbone, dataset) cluster at a time. The review that prompted this analysis asked for dataset fixed effects *and* leave-one-cluster-out validation; those two requests are jointly unsatisfiable, and satisfying them naively would have produced a table that looks right and means nothing. A cluster *is* a (backbone, dataset) pair, so holding one out removes the only cells that identify that dataset's fixed effect, its column is all-zero in the training fold, and the fold silently scores a model nobody specified. Three of the seven series (Electricity, Electricity-7, ILI) are covered by a single backbone each, so no other fold could identify them even in principle. We therefore run two schemes and label each for what it is. The leave-one-cluster-out ladder excludes dataset fixed effects and is genuinely out-of-sample; it is where ΔR^2 is computed. The variant that *carries* dataset fixed effects is reported as an in-sample fit (R^2 rising from $+0.642$ to $+0.711$, an incremental $+0.069$ from CKA, on 7 series and 31 cells with 12 parameters) and is explicitly not called validation.

The result. The registered prediction fires. Knowing the backbone buys $R^2_{\text{LOCO}}{=}+0.137$; adding horizon and $\log n$ takes it to $+0.201$; adding CKA *lowers* it to $+0.176$. $\Delta R^2{=}-0.025$, the median bootstrap draw is -0.083, and adding CKA fails to help in 69% of cluster-bootstrap draws. Two conventions in that sentence are worth stating. First, out-of-sample R^2 is scored against the **training-fold mean**, never the held-out fold's own mean: against its own mean a fold with little internal spread looks catastrophic and one with a lot looks good, so the aggregate would report fold composition rather than model quality. The intercept-only model scores exactly $+0.000$, which is the check that this convention is implemented as described. Second, mean absolute error is reported beside R^2 because the folds are unbalanced (cluster sizes run from 1 to 5 cells), and MAE moves in the same direction ($+0.59$ pp worse with CKA in the model).

One disclosed deviation from the registration, and it is a dispersion measure, not the deciding statistic. The registration said the cluster bootstrap would be reported on ΔR^2 itself. It cannot

usefully be: an out-of-sample R^2 is unbounded below, so a draw that duplicates clusters and leaves a one-cell fold can score an arbitrarily large negative value, and the 2.5th percentile of ΔR^2 came out at -3.3, -1.7 and -32.5 on bootstrap seeds 0, 1 and 7 while the point estimate, the median and the share of non-helping draws moved in the third decimal. Quoting that percentile would have put a number in this paper that a reader re-running the script with another seed cannot reproduce; ΔMAE's upper bound is unstable for the same reason. Both therefore carry a median and an interquartile range, which are seed-stable, and the 95% interval is quoted on the one *linear* coefficient in the analysis, CKA's slope in M3, $+43$ pp per unit CKA with a cluster-bootstrap interval of $[-31, +107]$, whose bounds move by under 5 pp across seeds. The registered *deciding statistic*, the sign of ΔR^2, is unchanged.

What this does and does not establish. It does not establish that CKA is uninformative about fine-tuning in general; it establishes that on these 31 cells, CKA carries no incremental out-of-sample information about the value of encoder adaptation once backbone identity, horizon and training-set size are known, and that the part of Δ_{enc} that *is* predictable is predictable from protocol identity rather than from the representation-similarity measurement. That is a sharper statement than a wide slope interval, and it is the sense in which a post-treatment similarity statistic fails to substitute for running the intervention. It is also worth naming what the ladder shows about the alternative: the interaction model M4 collapses to $+0.019$, barely better than the intercept, so "CKA predicts Δ_{enc} but with a backbone-specific slope" is the worst performing account of these data, not a rescue of the pooled correlation.

A18 The Pre-Registered Positive Control, in Full

What was registered, and when. `results/positive_control/preregistration.json` was committed at a named revision before any run of this arm existed, and it fixed: task A, chosen from the run records as $\arg\max V_{\text{ridge}}$ over the seven gate-passing cells (Moirai-Small/ETTh2, h=192, V_{ridge}=+0.932); task B by SHA-256, with its generator (`scripts/make_conflicting_series.py`); the grid; the primary readout; *which checkpoint is primary*; *where CKA is measured*; two falsifiable validity conditions; and all three outcome sentences.

Task B, and why its construction matters. $x_j(t) = a_j\,(-1)^{\lfloor t/P \rfloor}\, s_j(t \bmod P) + \sigma_j \varepsilon$, with s_j a three-harmonic unit-variance periodic shape and P=192=h. The correct forecast is therefore the *negation* of the previous period, so the seasonal-naive prior (which is task A's own strongest admissible baseline rung) is maximally wrong on task B: seasonal-naive loses to its own negation by $200.9\times$. This is what makes learning B a genuine conflict with A rather than the addition of an unrelated skill, and it is the reason a null here is informative.

Where CKA is measured, and why it differs from the main matrix. On **task A's** inputs, not task B's. In the main matrix the two coincide, because a cell trains and is evaluated on one series; here they cannot, and measuring drift on the series being trained *on* would answer a different question than "did the representation supporting A change". This was the design choice we expected to be probed, so it is registered and stated rather than buried.

Both checkpoints, and the noise floor that makes them readable. The early-stopped state is primary, as registered; the final-epoch state is reported beside it as the maximal-destruction bound, because the flag selects on task B's validation loss and so can return a checkpoint from before damage to A has finished developing. **The registration's stated reason for that choice (comparability with the main matrix, "which early-stops the same way") is mistaken**: the 21 Moirai cells of the matrix pass no `--early-stopping` flag and are read at their final epoch (§3). We leave the registered choice and the registration's wording as they are and report both states, so the error is in a stated rationale and not in a result; the arm's own runs do restore the best selection-split epoch, which is what the primary column is. That epoch was often the last one, so the two states are frequently *identical*, and then the gap between their two retention numbers is not redundancy but a direct read of the evaluator's own sampling noise, since Moirai's forecast is the median of 20 samples. Over the 17 identical-state pairs that gap is at most 0.48 pp and 0.20 pp on average. **Every retention number in this arm should be read against that floor**, which is why the emitter computes it across all such pairs rather than asserting a tolerance.

The validity conditions, both of which hold. (i) Task B must actually be learned; destruction without learning is weight thrashing, not forgetting. It was, in **21 of 21** runs, by at least $+56.6\%$ against a registered 20% threshold. (ii) The frozen encoder should preserve task A; if it does not, the damage is not in the encoder. It does: retention stays inside $+5\%$ at all three completed rungs ($+1.3 \pm 0.4$, $+4.2 \pm 2.2$, $+3.4 \pm 9.4$), against a destruction threshold of $+50\%$.

The outcome, quoted from the registration. Destruction was not achieved: the registered rule was retention$_A \geq 50\%$ in a majority of seeds at some rung under condition B, and no rung meets it: the per-rung maxima are -7.1, $+26.7$, $+4.9$ and -20.1. The registered sentence for this outcome reads: "the positive control failed to construct… deliberately adversarial fine-tuning did not destroy the capability – itself a finding about TSFM robustness, and a reason the main matrix's null is unsurprising." Across all 21 runs Spearman ρ between CKA on task A's inputs and retention is $+0.466$ (p=0.03, n=21): a positive coefficient is the wrong sign for a preservation heuristic, so the ordering fails here as it does retrospectively.

The declared extension, and the three runs that diverged. The stopping rule permitted one extension to lr=10^{-1} if destruction was not achieved at 10^{-2}, and then a stop. Condition B completed all three seeds and supplies the ladder's sharpest row. Condition D diverged in 3 of 3: with the encoder frozen every gradient lands on the decoder, and its likelihood head leaves the domain its distribution requires: two seeds on the negative binomial's `total_count`, one on a categorical's `logits`. **We do not re-run them.** Guarding the evaluation call and retrying would be re-running the declared extension after seeing its outcome, which is the thing a pre-registration exists to prevent; the divergence is reported as the result for that rung. It is also not load-bearing: condition D's role is to localise damage in the encoder, and there is no damage anywhere in the ladder to localise.

What this arm cannot show, as registered: that the capability at stake was valuable against a strong baseline (task A beats seasonal-naive by only $+0.081$, and does not clear the full admissible ladder, so this tests *detection* of a retention loss rather than the value of what is lost); that anything here generalises beyond one backbone, one series and one synthetic task B; or that a sign-inverting target task resembles a deployment scenario. It also cannot show that a retention loss is impossible on these benchmarks, only that we could not construct one, having tried four learning-rate rungs spanning three orders of magnitude with a task built to conflict.

Table 19: **Paired seed-level inference on encoder adaptation.** $\Delta_{\text{enc}} = L_{\text{full}} - L_{\text{frozen}}$ per seed, in percent of the cell's zero-shot reference, so a positive value means allowing the encoder to adapt made the cell *worse*. B and D share a seed, an initialisation and a window sample, so this contrast is paired by construction. d is paired Cohen's d. The SEM column was dropped when q was added (nine columns overrun the textwidth and the interval states the same dispersion), so `sem` lives in `results/paired_inference.json`, on the body's convention (§3). Each row names its own n, which is not constant down the column (Table 7). q **is the decision column, adjusted for the 31 tests**: Benjamini–Hochberg within this contrast, with the three-way call (freeze-decisive, adapt-decisive, inconclusive) read off $q<0.05$ and the sign of Δ_{enc}, and bolded where decisive. BH runs on p_t, not p_{exact}, because 24 cells have an exact floor of 0.25 and a step-up procedure there would reject nothing at any effect size. **Two tests, because neither alone is adequate at these seed counts.** p_{exact} assumes nothing, but its resolution is fixed by N: the 24 cells at $N{=}3$ cannot reach $p < 0.05$ at any effect size, their smallest attainable value being 0.25. p_t and the interval do resolve there, and pay for it with a normality assumption on three points that cannot be checked. Both are shown so that a reading resting on that assumption is visible as one; the text states why no bootstrap interval is reported. In the two summary rows the last column is *not* an effect size: it gives the number of cells whose mean Δ_{enc} is positive, out of the cells in that row. † marks cells clearing the gate threshold under at least one admissible baseline of the ladder (Appendix A11). ‡ marks the Chronos cells, whose per-condition forgetting is confounded by the head/decoder mismatch; their Δ_{enc} is unaffected.

Cell	N	Δ_{enc}	95% CI (t)	p_t	q	p_{exact}	d
Chronos / ETTh1 (h=24, n=8000)†‡	3	−39.2	[−98.9, +20.5]	0.106	0.173	0.250	−1.63
Chronos / ETTh2 (h=24, n=8000)†‡	3	−12.9	[−15.5, −10.4]	0.002	**0.016**	0.250	−12.57
Chronos / Weather (h=24, n=8000)†‡	3	−21.6	[−35.5, −7.6]	0.022	0.085	0.250	−3.84
Moirai-B / ETTh2 (h=192, n=1000)†	3	−20.2	[−24.9, −15.5]	0.003	**0.018**	0.250	−10.66
Moirai-B / ETTh2 (h=96, n=1000)†	3	−22.3	[−34.7, −10.0]	0.016	0.071	0.250	−4.50
Moirai-B / ETTh1 (h=96, n=1000)†	3	+37.5	[+2.1, +72.9]	0.045	0.099	0.250	+2.63
Moirai-B / ETTm2 (h=192, n=1000)†	3	+1.3	[−13.4, +16.0]	0.740	0.740	1.000	+0.22
Moirai-B / ETTm2 (h=96, n=1000)†	3	+7.0	[−3.2, +17.3]	0.098	0.173	0.250	+1.70
Moirai-L / ETTh2 (h=96, n=500)†	3	−20.8	[−26.6, −15.0]	0.004	**0.022**	0.250	−8.87
Moirai-L / ETTh1 (h=96, n=1000)†	3	+28.2	[+7.3, +49.1]	0.028	0.098	0.250	+3.35
Moirai-S / ETTh1 (h=96, n=1000)†	5	+22.1	[+15.6, +28.6]	0.001	**0.011**	0.062	+4.21
Moirai-S / ETTh2 (h=192, n=500)†	10	+6.1	[−1.5, +13.7]	0.101	0.173	0.100	+0.58
Moirai-S / ETTh2 (h=96, n=500)†	10	+2.8	[−5.6, +11.2]	0.472	0.522	0.557	+0.24
Moirai-S / ETTm2 (h=192, n=1000)†	5	+4.7	[−4.2, +13.6]	0.217	0.306	0.312	+0.65
Moirai-S / ETTm2 (h=96, n=1000)†	5	+2.6	[−6.9, +12.1]	0.495	0.529	0.562	+0.34
Moirai-S / Electricity7 (h=96, n=1000)†	3	−3.2	[−9.9, +3.4]	0.170	0.251	0.250	−1.22
Moirai-S / Weather (h=96, n=1000)†	3	+53.1	[+9.2, +97.0]	0.035	0.099	0.250	+3.00
TimesFM / ETTh1 (h=24, n=1000)†	3	−1.4	[−13.5, +10.6]	0.658	0.680	0.750	−0.30
TimesFM / ETTh2 (h=24, n=1000)†	3	−7.1	[−20.5, +6.2]	0.148	0.230	0.250	−1.33
TimesFM / Weather (h=24, n=1000)†	3	−1.6	[−8.1, +4.8]	0.387	0.445	0.500	−0.63
Chronos / Electricity (h=24, n=8000)‡	3	−2.7	[−10.7, +5.4]	0.287	0.358	0.500	−0.83
Chronos / ETTm2 (h=24, n=8000)‡	3	−29.5	[−57.1, −1.9]	0.044	0.099	0.250	−2.66
Moirai-B / ETTh1 (h=192, n=1000)	3	+47.0	[−12.4, +106.3]	0.077	0.158	0.250	+1.97
Moirai-B / Weather (h=192, n=1000)	3	+12.5	[−25.1, +50.1]	0.288	0.358	0.500	+0.83
Moirai-B / Weather (h=96, n=1000)	3	+9.4	[+1.0, +17.8]	0.040	0.099	0.250	+2.78
Moirai-L / Weather (h=96, n=1000)	3	+12.0	[−20.3, +44.3]	0.251	0.338	0.250	+0.92
Moirai-S / ETTh1 (h=192, n=1000)	5	+13.5	[+8.8, +18.2]	0.001	**0.014**	0.062	+3.53
Moirai-S / ILI (h=24, n=452)	10	−38.6	[−41.6, −35.6]	0.000	**0.000**	0.002	−9.15
Moirai-S / Electricity7 (h=192, n=1000)	3	−2.6	[−11.7, +6.4]	0.338	0.403	0.500	−0.72
Moirai-S / Weather (h=192, n=1000)	3	+62.1	[+3.6, +120.6]	0.045	0.099	0.250	+2.64
TimesFM / ETTm2 (h=24, n=1000)	3	−7.7	[−18.9, +3.5]	0.097	0.173	0.250	−1.71
Mean of cell means, 95% CI from resampling (backbone, dataset) clusters:							
20 value-cells / 11 clusters		+0.7	[−11.1, +14.1]				10/20
all 31 cells / 15 clusters		+2.9	[−12.7, +15.1]				16/31

Table 20: **What each surviving cell's own seed budget could detect.** The 7 cells clearing $V_{\text{ridge}} \geq 0.20$, with the minimum detectable effect implied by their own dispersion: MDE $= t_{1-\alpha/2,n-1}\,\text{sd}/\sqrt{n}$, the smallest $|\Delta_{\text{enc}}|$ a two-sided paired t would resolve at that n. $n^\star$ is the paired seed count at which the *observed* effect would resolve, holding the observed sd fixed; it is therefore a post-hoc estimate and not a design calculation. **Two columns, because a single test and a decisive call cost different budgets.** $n^\star$ is computed at an uncorrected $\alpha{=}0.05$; $n^\star_{\text{adj}}$ is computed at α/m with $m{=}31$ tests, which bounds the Benjamini–Hochberg threshold the call in Table 19 actually uses from below and so bounds the budget from above. Reporting only $n^\star$ would say Moirai-Base/ETTh2 $h{=}192$ is already resolved at $n{=}3$, which its own $q{=}0.018$ happens to agree with, and would say the same of $h{=}96$, whose q is 0.071. **The 4 starred rows are the point of the table**: they need more than our feasibility cap of 20 paired seeds ($\approx$20 GPU-hours per cell at two fine-tunes per seed), so we report them as *unresolvable under this design*, with the number, rather than as effects that are absent.

Cell	V_{ridge}	n	Δ_{enc}	sd	MDE	$n^\star$	$n^\star_{\text{adj}}$
Moirai-S / ETTh2 ($h{=}192$, $n{=}500$)	$+0.932$	10	$+6.1$	10.63	7.60	14	$36^\star$
Moirai-B / ETTh2 ($h{=}192$, $n{=}1000$)	$+0.925$	3	-20.2	1.90	4.71	2	4
Moirai-L / ETTh2 ($h{=}96$, $n{=}500$)	$+0.875$	3	-20.8	2.35	5.83	3	4
Moirai-S / ETTh2 ($h{=}96$, $n{=}500$)	$+0.874$	10	$+2.8$	11.78	8.43	71	$182^\star$
Moirai-B / ETTh2 ($h{=}96$, $n{=}1000$)	$+0.777$	3	-22.3	4.97	12.33	3	5
Moirai-S / ETTm2 ($h{=}192$, $n{=}1000$)	$+0.352$	5	$+4.7$	7.16	8.89	12	$29^\star$
Moirai-B / ETTm2 ($h{=}192$, $n{=}1000$)	$+0.211$	3	$+1.3$	5.90	14.66	83	$212^\star$

Table 21: Is the pooled CKA correlation a backbone artefact? The same statistic (Spearman ρ of CKA against held-out B$-$D, 10,000-resample bootstrap CI) computed pooled and then stratified by backbone, at both resampling levels. "Cl." is the number of (backbone, dataset) clusters, which is the effective sample size for the clustered interval and is far below n. Values are emitted by `scripts/clustered_inference.py`. The pooled row is reported for audit, not read as evidence; the two small arms carry essentially no information at their cell counts, and for them the two levels coincide because every cell is already its own cluster.

Grouping	n	Cl.	CKA range	ρ	95% CI over cells	95% CI over clusters
Pooled (against our rule)	31	15	0.09–0.97	$+0.567$	$[+0.265, +0.759]$ excl. 0	$[+0.239, +0.828]$ excl. 0
Moirai only	22	6	0.36–0.97	$+0.168$	$[-0.328, +0.598]$ incl. 0	$[-0.469, +0.717]$ incl. 0
Chronos only	5	5	0.09–0.23	$+0.100$	$[-1.000, +1.000]$ incl. 0	$[-1.000, +1.000]$ incl. 0
TimesFM only	4	4	0.25–0.56	$+0.400$	$[-0.778, +1.000]$ incl. 0	$[-0.778, +1.000]$ incl. 0

Table 22: **The within-backbone dissociation, one capacity, eight cells.** Every Moirai-Base cell, ordered by CKA (most drifted first). "Full FT" and "Frozen" are percent change in held-out MSE against zero-shot, so negative is better and positive is task-level forgetting; Δ_{enc} is the manipulated contrast between them. Gate values marked ^{f} fail the 0.20 screen. Reading down the CKA column, the outcome reverses sign: the two least drifted cells are the only two that degrade. Emitted by `scripts/emit_dissociation.py`, which asserts both of those claims against the run records and exits non-zero if they stop holding.

Cell (h)	CKA	Gate	Full FT	Frozen	Δ_{enc}
ETTh2 (192)	0.396	$+0.925$	-36.3%	-16.1%	-20.2
ETTh2 (96)	0.460	$+0.777$	-31.4%	-9.1%	-22.3
ETTm2 (192)	0.488	$+0.211$	-8.4%	-9.7%	$+1.3$
Weather (192)	0.509	-0.633^{f}	-18.1%	-30.6%	$+12.5$
ETTm2 (96)	0.510	-0.257^{f}	-13.1%	-20.1%	$+7.0$
Weather (96)	0.567	-0.823^{f}	-11.7%	-21.1%	$+9.4$
ETTh1 (192)	0.669	-0.600^{f}	$+39.5\%$	-7.5%	$+47.0$
ETTh1 (96)	0.755	-0.312^{f}	$+22.8\%$	-14.7%	$+37.5$

Table 23: Within-backbone slope of held-out B−D on CKA, in percentage points per unit of CKA, with backbone identity absorbed as a fixed effect rather than by subsetting. Intervals are 10,000 cluster-bootstrap draws over the 15 (backbone, dataset) clusters of 31 cells; draws in which the fixed effects are unidentified or the design is rank-deficient are discarded, and the kept count is reported. Emitted by `scripts/clustered_inference.py` from `scripts/cka_fixed_effects.py`.

Specification	b_1 (pp per unit CKA)	95% CI over clusters	Draws kept
No backbone term	$+47.71$	—	—
CKA + backbone FE	$+39.08$	$[-15.37, +98.50]$ includes 0	9895
CKA + backbone FE + CKA×backbone	$+39.91$	$[-18.95, +109.71]$ includes 0	8522

Table 24: **Does CKA add out-of-sample predictive information about Δ_{enc} beyond backbone identity?** Nested models over all 31 scored cells, validated by leave-one-cluster-out across the 15 (backbone, dataset) clusters. R^2_{LOCO} is scored **against the training-fold mean**, never the held-out fold's own mean, so a positive value means the model beats predicting the other clusters' average; MAE is in percentage points of zero-shot loss. **The deciding statistic and this table's interpretation were fixed before the models were fitted** (`results/preregister_loco.json`): $\Delta R^2 = R^2_{\text{LOCO}}(\text{M3}) - R^2_{\text{LOCO}}(\text{M2}) = -0.025$, median bootstrap draw -0.083, IQR $[-0.227, +0.043]$, and adding CKA fails to help in 69% of draws; $\Delta\text{MAE} = +0.59$ pp, IQR $[-0.37, +2.98]$. **Interquartile ranges, not 95% intervals, are quoted for both**: an out-of-sample R^2 is unbounded below and a degenerate bootstrap fold can blow up either statistic, so their extreme percentiles are not reproducible: ΔR^2's 2.5th percentile moved between -1.7 and -32.5 across bootstrap seeds, and ΔMAE's 97.5th between $+11$ and $+19$, while the point estimates, medians and sign share moved in the third decimal. **The 95% interval is carried instead by the one linear coefficient here**, CKA's slope in M3: $+43$ pp per unit CKA, cluster-bootstrap CI $[-31, +107]$, which covers zero, so the slope's sign is not even determined, let alone its size. That interval is quoted to whole percentage points because its bounds move by under 5 pp across bootstrap seeds. The pre-registered deciding statistic (the *sign* of ΔR^2) is unchanged; only the dispersion measure beside it differs from the registration, which says so. Dataset identity is deliberately *absent* from the ladder: a cluster is a (backbone, dataset) pair, so holding one out removes the only cells that identify that dataset's fixed effect and the fold would silently score a model nobody specified: three of the 7 series are covered by a single backbone, so no other fold could identify them even in principle. The in-sample fit *with* dataset fixed effects is reported in the text and is not validation.

Model	Predictors	R^2_{LOCO}	MAE
M0	intercept only	$+0.000$	20.05
M1	backbone FE	$+0.137$	18.77
M2	backbone FE + horizon + $\log n$	$+0.201$	18.78
M3	M2 + CKA	$+0.176$	19.37
M4	M2 + CKA × backbone	$+0.019$	20.78

Table 25: **The pre-registered positive control: 21 runs, no destruction.** Task A is Moirai-Small/ETTh2 h=192; task B is the anti-persistent synthetic series, fixed by SHA-256 before any run. "task B learned" is the MSE reduction on B's own held-out windows (registered validity threshold 20%). "retention on task A" is $100(\text{MSE}_A(\theta_{\text{FT}}) - \text{MSE}_A(\theta_{\text{ZS}}))/\text{MSE}_A(\theta_{\text{ZS}})$, so *positive is worse* and the registered destruction threshold is $+50\%$ in a majority of seeds under full fine-tuning. CKA is measured on **task A's** inputs. $\pm$ is the SEM over 3 seeds. † marks the single declared ladder extension. Read every retention number against the 0.48 pp evaluator noise floor derived in the text.

Diverged, not omitted: 3 run(s) at lr $= 0.1$, frozen encoder did not complete; training diverged and the run is reported as a divergence rather than a number (see text).

		task B	retention on task A (%)		CKA on	ℓ_2
lr	encoder	learned (%)	early-stopped	final epoch	task A	drift
0.0001	full fine-tune	$+99.0_{\pm 0.0}$	$-10.8_{\pm 1.9}$	$-10.7_{\pm 1.8}$	$0.796_{\pm 0.034}$	$7.63_{\pm 0.02}$
0.0001	frozen encoder	$+57.3_{\pm 0.6}$	$+1.3_{\pm 0.4}$	$+1.2_{\pm 0.3}$	$0.999_{\pm 0.000}$	$7.19_{\pm 0.06}$
0.001	full fine-tune	$+99.3_{\pm 0.0}$	$+8.6_{\pm 13.4}$	$+8.8_{\pm 13.6}$	$0.234_{\pm 0.032}$	$42.37_{\pm 0.85}$
0.001	frozen encoder	$+79.6_{\pm 0.7}$	$+4.2_{\pm 2.2}$	$+4.3_{\pm 2.4}$	$0.975_{\pm 0.003}$	$25.87_{\pm 0.49}$
0.01	full fine-tune	$+95.8_{\pm 1.2}$	$-13.4_{\pm 9.7}$	$-13.2_{\pm 9.6}$	$0.279_{\pm 0.101}$	$456.22_{\pm 16.10}$
0.01	frozen encoder	$+85.3_{\pm 1.0}$	$+3.4_{\pm 9.4}$	$+3.7_{\pm 9.3}$	$0.900_{\pm 0.055}$	$78.09_{\pm 3.38}$
$0.1^{\dagger}$	full fine-tune	$+84.1_{\pm 1.9}$	$-23.3_{\pm 2.0}$	$-23.3_{\pm 2.0}$	$0.063_{\pm 0.016}$	$2794.83_{\pm 40.24}$